

Large-scale AI-guided liver malignancy diagnosis: multicenter study and a single-arm trial

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Liver malignancies are frequently evaluated on contrast-enhanced computed tomography (CE-CT), but missed or delayed diagnoses remain a clinically important challenge in high-volume, real-world radiology workflows, highlighting the need for scalable diagnostic safety net approaches. To address this, we developed the Liver DiagnOsis Network (LiON), a CE-CT-based artificial intelligence (AI) system that supports flexible multiphase processing, clinical data integration and workflow-compatible liver malignancy diagnosis. LiON was trained on 6,443 patients and retrospectively validated across 22,251 patients from multicenter and real-world cohorts. LiON achieved high performance for malignancy diagnosis, with an area under the receiver operating characteristic curve (AUC) of 0.975 (95% confidence interval (CI): 0.971–0.979), and maintained robust performance in real-world cohorts and among patients with hepatic steatosis (AUC 0.971, 95% CI: 0.952–0.985) and cirrhosis (AUC 0.924, 95% CI: 0.901–0.946). We then conducted a single-arm trial in 10,333 patients in routine clinical practice, in which LiON functioned as an additional AI reader within the existing clinical workflow. The trial met its primary endpoint, defined as an AUC for malignancy diagnosis with the lower bound of the 95% CI exceeding 0.900, achieving an AUC of 0.952 (95% CI: 0.942–0.961). Secondary outcomes demonstrated that AI–human collaboration identified 51 previously overlooked lesions (15 malignancies) and triggered 37 amended radiology reports, 22 multidisciplinary team escalations and clinical management changes in a subset of patients. These findings suggest that AI, when deployed as a workflow-compatible diagnostic support, may help reduce missed or delayed diagnoses and guide clinical interventions. Nevertheless, further evidence from prospective comparative studies across diverse healthcare systems is warranted to assess effects on clinical outcomes. ClinicalTrials.gov identifier: [NCT07153783](https://clinicaltrials.gov/ct2/show/study/NCT07153783).

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Liver malignancies, spanning both primary cancer and metastasis, pose a substantial global health burden^{1,2}. Primary liver malignancy, predominantly hepatocellular carcinoma (HCC), ranks sixth in cancer incidence and is among the leading causes of cancer deaths worldwide, with cases projected to nearly double, reaching 1.52 million by 2050 (refs. 3,4). Hepatic metastases, however, are more common than primary liver malignancies². This high and growing burden underscores the critical need for timely and accurate diagnosis. Radiologically, these malignancies manifest as focal liver lesions (FLLs), for which accurate malignancy diagnosis strongly influences treatment and prognosis. In HCC, for instance, survival ranges from over 5 years in early stages to months in advanced stages⁵. Multiphase contrast-enhanced computed tomography (CE-CT) is widely used as a first-line modality for diagnosing liver malignancies and evaluating indeterminate FLLs, often initially detected by ultrasound or other non-contrast imaging, as reflected in international guidelines^{6–10}.

Nevertheless, interpreting CE-CT for FLLs represents a substantial workload in clinical practice. Comprehensive liver assessment is a mandatory component of all abdominal CE-CT examinations, making it time-consuming and requiring careful scrutiny to detect subtle abnormalities. Moreover, CE-CT faces several diagnostic challenges: reduced sensitivity for small lesions^{11,12}, difficulty in diagnosing metastases¹³ and challenging assessment in diffuse chronic liver diseases (for example, hepatic steatosis and cirrhosis)^{14,15}. These limitations increase the risk of missed or incorrect diagnoses, potentially delaying critical therapeutic interventions. Artificial intelligence (AI) has shown promise in diagnostic radiology^{16–19}, with recent AI advances in liver malignancy analysis showing superior performance in lesion delineation²⁰, staging accuracy²¹, patient triage^{22,23} and diagnostic support^{24,25}. However, most existing AI methods still lack validation in large-scale prospective studies^{26–28}, leaving it uncertain whether these systems maintain performance in heterogeneous real-world populations and, more critically, whether their clinical integration can tangibly improve the standard of care. This uncertainty also complicates the management of indeterminate lesions (for example, Liver Imaging Reporting and Data System (LI-RADS) 3/4 (ref. 29)), requiring a careful balance between the risks of delayed diagnosis and unnecessary invasive procedures¹².

To address these diagnostic gaps, we developed Liver DiagnOsis Network (LiON), an AI system for rapid automatic malignancy assessment and detection of eight major FLL subtypes. LiON accommodates flexible-phase CT inputs and can integrate clinical information when available. We validated LiON in a multistage study, including multicenter retrospective validation ($n = 22,251$) and a single-arm prospective study ($n = 10,333$). In routine radiological workflows, we assessed whether AI–human collaboration could identify primary misdiagnoses and overlooked malignancies, leading to modified treatment decisions (Extended Data Figs. 1 and 2).

Results

Study overview

We developed and validated LiON through a three-stage study (Extended Data Fig. 1b). Stage I trained LiON on 6,443 patients and evaluated it on internal and external cohorts (Cohort-IN, $n = 300$; Cohort-EX, $n = 4,071$), including a reader study with 11 radiologists and integration of clinical information. Stage II stress tested LiON in two high-stakes scenarios: early-stage HCC diagnosis on CE-CT benchmarked against contrast-enhanced magnetic resonance imaging (CE-MRI) interpretation (Cohort-HCC, $n = 150$) and hepatic metastasis diagnosis in patients with pancreatic cancer (Cohort-PDAC, $n = 1,511$). Stage III assessed real-world performance in consecutive retrospective cohorts (Cohorts-RWR1–RWR4, $n = 16,219$) and a prospective cohort (Cohort-RWP, $n = 10,333$) evaluating AI–human collaboration for clinical decision support. We mainly evaluated three capabilities: (1) malignancy diagnosis/detection, the primary task, which classifies

patients as malignant versus non-malignant (benign or no-lesion); (2) lesion subtype classification, which identifies detected lesions as one of eight subtypes; and (3) lesion screening, which distinguishes patients with any lesion (malignant and benign) from those with no lesions.

Stage I: foundational validation

LiON is an end-to-end CE-CT system for patient-level malignancy prediction and lesion-level segmentation of eight FLL subtypes, including malignant lesions—namely, HCC, intrahepatic cholangiocarcinoma (CCA), metastasis and other malignant lesions (treated lesions, sarcoma, etc.)—as well as benign lesions—namely, hemangioma, focal nodular hyperplasia (FNH), cysts and other benign lesions (abscess, calcification, inflammatory pseudotumors, etc.). The system supports optional delayed-phase input and clinical information integration when available (Extended Data Fig. 3). LiON was trained on patients from two major medical centers using supervised annotations of lesions, liver and perihepatic organs and was validated on multicenter datasets (cohort characteristics and annotation details provided in Table 1, Extended Data Table 1 and Methods).

Multicenter validation. To evaluate the generalizability of LiON, we retrospectively validated it on an internal cohort (Cohort-IN, $n = 300$) and an external cohort from five independent centers (Cohort-EX, $n = 4,071$), with diagnoses confirmed by pathology or ≥ 1 -year follow-up. For malignancy diagnosis, LiON achieved an area under the receiver operating characteristic curve (AUC) of 0.975 (95% confidence interval (CI): 0.971–0.979) across centers, with a sensitivity of 90.2% (95% CI: 89.0–91.5%) and a specificity of 97.6% (95% CI: 96.6–98.3%) in Cohort-EX (Fig. 1a,b). LiON also showed robust lesion screening performance in Cohort-EX (AUC 0.956, 95% CI: 0.948–0.963), with feature embeddings separating malignant from non-malignant cases (Fig. 1c–f).

We then evaluated LiON's flexible-phase architecture, which accommodates variable CT protocols through optional delayed-phase input. Compared to the self-configuring nnU-Net¹⁶ on Cohort-EX, LiON achieved superior lesion screening AUC (0.956 versus 0.926) and higher eight-class classification performance (micro-average AUC 0.952 versus 0.914) (Fig. 1g). Ablation analysis confirmed the diagnostic benefit of delayed-phase input, with significant AUC improvements for HCC, metastasis, hemangioma, FNH and cyst (all $P < 0.01$; Fig. 1h). A venous-phase-only variant remained robust despite modestly lower performance (screening AUC 0.949; eight-class micro-average AUC 0.938). Fairness analysis showed similar performance across clinical subgroups, including sex, etiology and liver background (Fig. 1i).

Reader study. We compared LiON with 11 radiologists (four senior, seven junior) in a three-round reader study on Cohort-IN ($n = 300$: 181 malignant, 75 benign and 44 no lesion), in which diagnoses were made using images alone, using images and clinical information and using images and clinical information with LiON assistance (Fig. 2a–d and Supplementary Fig. 4).

LiON achieved an AUC of 0.958 (95% CI: 0.932–0.979) for malignancy diagnosis, outperforming all individual radiologists in image-only interpretation (AUC range: 0.654–0.826; Fig. 2a). Compared to reading assisted by clinical information, LiON assistance improved radiologists' balanced accuracy by 6.6% and sensitivity by 11.5%, with junior radiologists reaching accuracy similar to unassisted senior radiologists. These gains were consistent across clinically relevant subgroups stratified by Barcelona Clinic Liver Cancer (BCLC) stage, extrahepatic malignancy status and cirrhosis status (Fig. 2e–g). AI assistance also reduced interpretation time by 27.1% (95% CI: 21.9–32.3%), with junior radiologists benefiting most (Fig. 2h). Lesion-wise classification remained high for common subtypes, including HCC, hemangioma and cysts (Fig. 2i).

Table 1 | Cohort characteristics

Stages		Stage I: foundation validation				Stage II: extended validation				Stage III: real-world validation				Real-world prospective			
Study phase	Training	Internal validation	External validation			Extended validation			Real-world retrospective					Real-world prospective			
	Internal	External				Internal	External	Internal	External				Internal	Internal			
Cohort abbreviation															Cohort-RWP		
Institute	SH-CMU	FAH-ZU	SH-CMU, FAH-ZU	HCH	SAH-BMC	SYSUCC	YYH	CSPH	SH-CMU	SIIPD	FAH-ZU	Cohort-RWR1	Cohort-RWR2	Cohort-RWR3	SH-CMU	SH-CMU	SH-CMU
Total cohort size	2,657	3,786	300	2,401	740	654	219	57	150	1,511	4,597	10,733	507	382	10,333		
Diagnosis category	Malignant (53.0%)	2,260 (59.7%)	181 (60.3%)	1,354 (56.4%)	264 (35.7%)	516 (78.9%)	170 (77.6%)	33 (57.9%)	100 (66.7%)	184 (12.2%)	241 (5.2%)	515 (4.8%)	1 (0.2%)	4 (1.0%)	537 (5.2%)		
	Benign	1,226 (32.4%)	75 (25.0%)	874 (36.4%)	96 (13.0%)	138 (21.1%)	49 (22.4%)	17 (29.8%)	50 (33.3%)	—	2,101 (45.7%)	4,363 (40.7%)	—	—	3,655 (35.4%)		
	No lesion	300 (7.9%)	44 (14.7%)	173 (7.2%)	380 (51.4%)	0 (0.0%)	0 (0.0%)	7 (12.3%)	0 (0.0%)	—	2,255 (49.1%)	5,855 (54.6%)	—	—	6,141 (59.4%)		
Lesion characteristics (patients / lesions / size (mean ± s.d. cm ³))	HCC	845 / 1,065 / 41.3 ± 123.1	120 / 155 / 70.9 ± 151.7	407 (17.0%)	4 (0.5%)	458 (70.0%)	170 (77.6%)	26 (45.6%)	100 (66.7%)	—	30 (0.7%)	38 (0.4%)	—	—	—		
	CCA	133 / 249 / 82.8 ± 164.7	18 / 18 / 73.3 ± 60.5	148 (6.2%)	9 (1.2%)	30 (4.6%)	0	7 (12.3%)	0	—	22 (0.5%)	20 (0.2%)	—	—	—		
	Metastasis	373 / 2,352 / 22.5 ± 146.1	27 / 97 / 5.8 ± 13.0	774 (32.2%)	251 (33.9%)	28 (4.3%)	0	0	0	184 (12.2%)	184 (4.0%)	439 (4.0%)	—	—	—		
	Hemangioma	306 / 687 / 140.7 ± 378.5	38 / 98 / 88.7 ± 188.4	518 (21.6%)	103 (13.9%)	42 (6.4%)	56 (25.6%)	9 (15.8%)	53 (35.3%)	—	417 (9.1%)	1,016 (9.4%)	—	—	—		
	FNH	77 / 89 / 63.6 ± 96.6	15 / 15 / 90.8 ± 179.7	8 (0.3%)	0	0	0	0	3 (2.0%)	—	10 (0.2%)	20 (0.2%)	—	—	—		
	Cyst	636 / 2,101 / 2.2 ± 23.5	98 / 228 / 2.2 ± 11.5	1,012 (42.1%)	69 (9.3%)	276 (42.2%)	55 (25.1%)	18 (31.6%)	101 (67.3%)	—	1,757 (38.2%)	3,504 (32.6%)	—	—	—		
	Other malignant	177 / 239 / 203.7 ± 346.8	20 / 21 / 166.0 ± 197.5	25 (1.0%)	0	0	0	0	0	—	14 (0.3%)	24 (0.2%)	—	—	—		
	Other benign	581 / 1,502 / 39.0 ± 98.2	47 / 59 / 22.8 ± 102.2	131 (5.5%)	14 (1.9%)	28 (4.3%)	12 (5.5%)	2 (3.5%)	2 (1.3%)	—	507 (11.0%)	1,090 (10.2%)	—	—	—		
	All malignant	1,409 / 3,869 / 61.4 ± 219.2	181 / 291 / 56.2 ± 130.5	1,354 (56.4%)	264 (35.7%)	516 (78.9%)	170 (77.6%)	33 (57.9%)	100 (66.7%)	—	241 (5.2%)	515 (4.8%)	—	—	537 (5.2%)		

Table 1 (continued) | Cohort characteristics

Stages	Stage I: foundation validation			Stage II: extended validation			Stage III: real-world validation								
All benign	1,386 / 4,379 / 378 ± 168.8	2,461 / 9,132 / 21.0 ± 90.9	75 / 400 / 29.7 ± 113.1	874 (36.4%) / 2,390 (99.5%) / 11 (0.5%)	96 (13.0%) / 703 (95%) / 37 (5.0%) / 12 (1.8%) / 11 (5.0%) / 4 (7.0%) / 53 (93.0%)	138 (21.1%) / 642 (98.2%) / 487 (74.5%) / 167 (25.2%) / 74 (33.8%) / 25 (43.9%)	49 (22.4%) / 208 (95.0%) / 145 (66.2%) / 167 (25.2%) / 74 (33.8%) / 25 (43.9%)	17 (29.8%) / 642 (98.2%) / 32 (56.1%) / 167 (25.2%) / 74 (33.8%) / 25 (43.9%)	50 (33.3%) / 112 (74.7%) / 109 (72.7%) / 41 (27.3%)	—	2,101 (45.7%) / 4,593 (99.9%) / 4 (0.1%)	4,363 (40.7%) / 7,248 (67.5%) / 3,485 (32.5%)	—	—	3,655 (35.4%)
	3.7 ± 6.0	4.0 ± 5.1	2.7 ± 2.8	—	—	—	—	—	—	—	—	—	—	—	—
	1,069 (40.2%) / 1,588 (59.8%)	1,498 (39.6%) / 2,288 (60.4%)	199 (66.3%) / 101 (33.7%)	2,390 (99.5%) / 11 (0.5%)	37 (5.0%) / 703 (95%) / 37 (5.0%) / 12 (1.8%) / 11 (5.0%) / 4 (7.0%) / 53 (93.0%)	12 (1.8%) / 642 (98.2%) / 487 (74.5%) / 167 (25.2%) / 74 (33.8%) / 25 (43.9%)	11 (5.0%) / 208 (95.0%) / 145 (66.2%) / 167 (25.2%) / 74 (33.8%) / 25 (43.9%)	4 (7.0%) / 53 (93.0%) / 32 (56.1%) / 167 (25.2%) / 74 (33.8%) / 25 (43.9%)	38 (25.3%) / 112 (74.7%) / 109 (72.7%) / 41 (27.3%)	1,511 (100.0%) / 0 (0.0%)	4,593 (99.9%) / 4 (0.1%)	7,248 (67.5%) / 3,485 (32.5%)	189 (37.3%) / 318 (62.7%)	128 (33.5%) / 254 (66.5%)	3,793 (36.7%) / 6,540 (63.3%)
Sex (male / female)	1,563 (58.8%) / 1,094 (41.2%)	2,200 (58.1%) / 1,586 (41.9%)	190 (63.3%) / 110 (36.7%)	1,447 (60.3%) / 954 (39.7%)	438 (59.2%) / 302 (40.8%)	487 (74.5%) / 167 (25.2%) / 74 (33.8%) / 25 (43.9%)	145 (66.2%) / 167 (25.2%) / 74 (33.8%) / 25 (43.9%)	32 (56.1%) / 167 (25.2%) / 74 (33.8%) / 25 (43.9%)	109 (72.7%) / 41 (27.3%)	899 (59.5%) / 612 (40.5%)	2,497 (54.3%) / 2,100 (45.7%)	4,795 (44.7%) / 5,938 (55.3%)	178 (35.1%) / 329 (64.9%)	224 (58.6%) / 158 (41.4%)	4,539 (43.9%) / 5,794 (56.1%)
Ages	55.0 ± 13.8	55.6 ± 13.2	59.9 ± 14.2	59.8 ± 10.9	56.6 ± 15.2	54.7 ± 12.6	57.5 ± 12.0	57.4 ± 16.1	56.6 ± 10.8	61.0 ± 10.8	58.2 ± 14.0	57.0 ± 13.8	53.4 ± 16.7	58.9 ± 11.4	57.2 ± 14.2

A dash (—) indicates not applicable.

Integration of clinical information. To assess whether clinical information could enhance imaging-based diagnosis, we developed a two-stage multimodal framework integrating LiON imaging predictions with 19 clinical attributes (demography, blood biomarkers and medical history) when available (Extended Data Fig. 3f). Trained on 1,646 internal patients with complete clinical information, the multimodal model outperformed image-only and clinical-only models. In Cohort-IN, it improved metastasis diagnosis compared to imaging alone, increasing AUC from 0.937 (95% CI: 0.873–0.983) to 0.983 (95% CI: 0.966–0.996) and sensitivity from 85.2% (95% CI: 70.8–96.9%) to 96.3% (95% CI: 87.5–100.0%). It also enhanced overall malignancy diagnosis (AUC 0.969, 95% CI: 0.949–0.985) and subtype classification performance (Extended Data Fig. 4 and Supplementary Fig. 8). Shapley Additive exPlanations (SHAP) analysis identified clinically plausible predictors, including α -fetoprotein (AFP) for HCC, carbohydrate antigen 19-9 (CA 19-9) for CCA and extrahepatic malignancy history and carcinoembryonic antigen (CEA) for metastasis (Extended Data Fig. 4 and Supplementary Fig. 19).

Stage II: extended validation

For stress testing, we evaluated LiON in two high-stakes clinical scenarios: early-stage HCC diagnosis on CE-CT benchmarked against CE-MRI interpretation and hepatic metastasis diagnosis in patients with pancreatic ductal adenocarcinoma (PDAC).

Diagnosis of early-stage HCC compared to MRI. Early-stage HCC diagnosis on CE-CT is challenging, particularly in anatomically complex locations (for example, inferior hepatic edges and vessel-rich left lobe areas), whereas CE-MRI utilizing gadoteric acid (Gd-EOB-DTPA) offers higher sensitivity and is recommended as a preferred imaging approach^{30–33}. We collected a paired-imaging cohort (Cohort-HCC, $n=150$) comprising 100 patients with early-stage HCC (BCLC 0-A) and 50 patients with benign lesions, all smaller than 3 cm and confirmed by pathology or ≥ 1 -year follow-up. Using CE-CT inputs alone, LiON was compared with two radiologists interpreting the corresponding CE-MRI scans (Fig. 3a–d). LiON achieved an AUC of 0.972 (95% CI: 0.943–0.994) and a sensitivity of 85.0% (95% CI: 77.7–91.3%). Although its sensitivity was lower than that of the senior radiologist on CE-MRI, LiON achieved high specificity and overall diagnostic performance that approached MRI-based radiologist interpretation in this paired cohort (Fig. 3a,b). These findings suggest that LiON may help support early-stage HCC assessment on CE-CT while MRI remains the preferred modality when available and clinically appropriate.

Diagnosis of hepatic metastases from pancreatic cancer. Given that CE-CT is the primary imaging modality for PDAC staging and resectability assessment^{34,35}, we analyzed a cohort (Cohort-PDAC, $n=1,511$) of patients with PDAC at different disease stages, including 269 with metastatic pancreatic cancer (MPC), of whom 184 had hepatic metastases. Metastatic status was confirmed through histopathological biopsy or multidisciplinary team (MDT) assessment based on clinical evidence (for example, positron emission tomography/computed tomography (PET/CT) and follow-up imaging). LiON demonstrated high patient-level performance for hepatic metastasis diagnosis, achieving an AUC of 0.979 (95% CI: 0.972–0.985), a sensitivity of 93.5% (95% CI: 90.0–96.9%) and a specificity of 94.5% (95% CI: 93.2–95.7%) (Fig. 3e–g). LiON also differentiated patients with hepatic involvement from those with only extrahepatic metastases (AUC 0.981, 95% CI: 0.956–0.997), showing its potential for PDAC staging.

Stage III: real-world validation

Finally, to assess the robustness of LiON in routine clinical practice, we further validated it in large-scale consecutive CE-CT cohorts from two major medical centers (Shengjing Hospital of China Medical University (SH-CMU) and The First Affiliated Hospital, Zhejiang University School

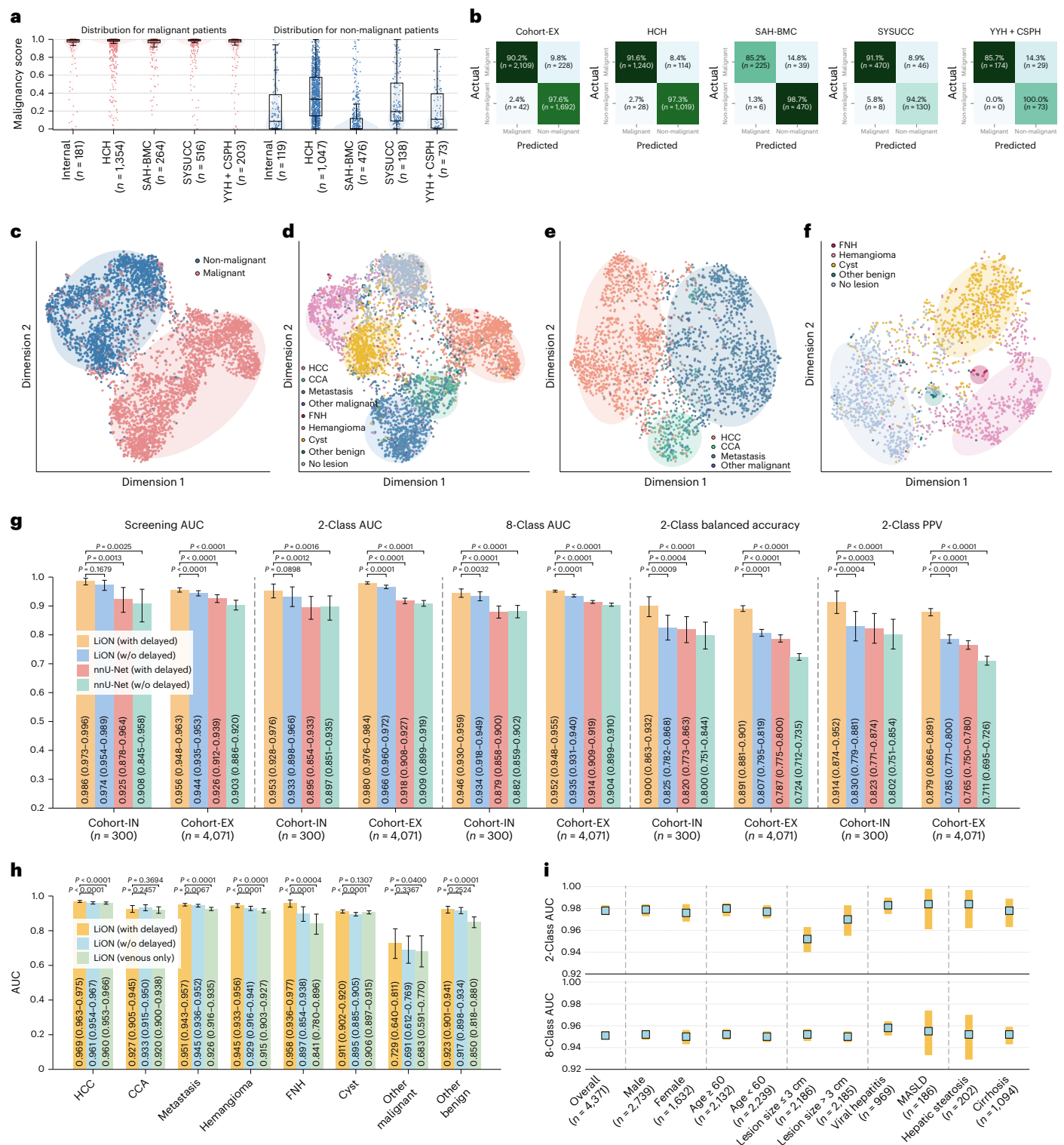


Fig. 1 | Multicenter performance evaluation. **a**, Distribution of malignancy scores for patients with and without malignancy. Violin plots show probability density distributions, with box plots displaying median, interquartile range (IQR) (box) and $1.5 \times$ IQR whiskers. Individual data points are overlaid as dots. **b**, Binary confusion matrices for LiON in malignancy diagnosis on Cohort-EX. **c–f**, t-SNE visualization of patient-level features extracted by the LiON classifier: binary classification (**c**); overall subtype classification (**d**); malignant subtypes (**e**); and non-malignant subtypes (**f**). Coloring reflects the ground truth labels. For

patients with multiple lesions, labels are assigned according to the most clinically relevant lesion. **g–i**, Detection performance comparison: overall metrics by center ($n = 4,371$ patients) (**g**); subtype-specific AUC (**h**); and clinical subgroup fairness analysis (**i**). n denotes the number of patients included in each cohort or subgroup. In **g–i**, the metrics are reported with 95% CIs from 2,000 bootstrap replications. Error bars represent 95% CIs. In **g** and **h**, P-values were calculated using two-sided DeLong tests for binary AUCs and two-sided paired permutation tests with 10,000 permutations for all other metrics.

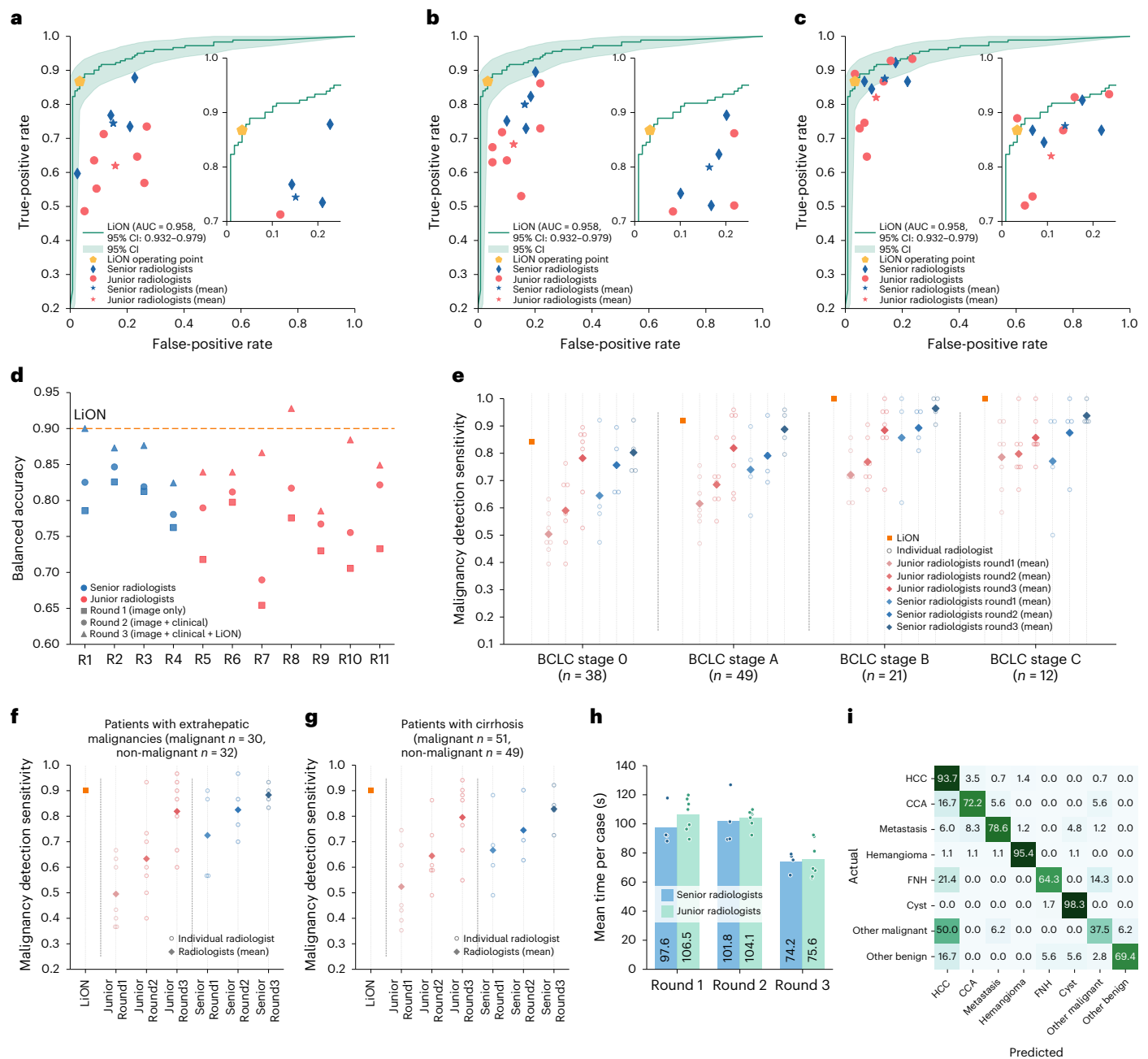


Fig. 2 | Comparative analysis of LiON and radiologists. **a–c**, Receiver operating characteristic curves for malignancy diagnosis across rounds: image-only diagnosis (round 1) (**a**); diagnosis using imaging and clinical information (round 2) (**b**); and diagnosis using imaging and clinical information with LiON assistance (round 3) (**c**). **d**, Impact of LiON assistance on reader performance in malignancy detection. **e–g**, Stratified subgroup analyses of malignancy

detection sensitivity in patients with HCC across BCLC stages (**e**), patients with extrahepatic malignancies (**f**) and patients with cirrhosis (**g**). **h**, Comparison of mean interpretation times across reader groups, with $n = 4$ senior readers and $n = 7$ junior readers. **i**, Confusion matrix of lesion-wise classification in recalled samples. In **a–c**, the metrics are reported with 95% CIs from 2,000 bootstrap replications. Error bars represent 95% CIs. R, radiologist.

of Medicine (FAH-ZU)), reflecting heterogeneous real-world populations dominated by normal and benign findings (Extended Data Fig. 5).

The real-world validation addressed two aims: (1) retrospective evaluation of LiON as a clinical assistance tool, identifying scenarios where AI predictions are reliable (Cohort-RWR1 and Cohort-RWR2), including malignancy screening in high-risk populations (Cohort-RWR3 and Cohort-RWR4), and (2) a single-arm trial of AI–human collaboration in actual clinical practice (Cohort-RWP). This study assessed AI–human collaboration, examining AI’s complementary role in detecting and diagnosing potentially overlooked or misdiagnosed lesions during routine clinical workflows. We also evaluated AI’s synergistic

value as a reliable second opinion when radiologists encountered diagnostic uncertainty.

Real-world retrospective study. For the real-world retrospective study, we deployed LiON across four cohorts. The primary validation was conducted on 15,330 consecutive patients from two centers (Cohort-RWR1, $n = 4,597$; Cohort-RWR2, $n = 10,733$). Additionally, we established two screening cohorts from SH-CMU: an oncology follow-up cohort (Cohort-RWR3, $n = 507$) and a high-risk cirrhotic cohort (Cohort-RWR4, $n = 382$), and these patients had received an initial non-malignant clinical report for the index CE-CT examination.

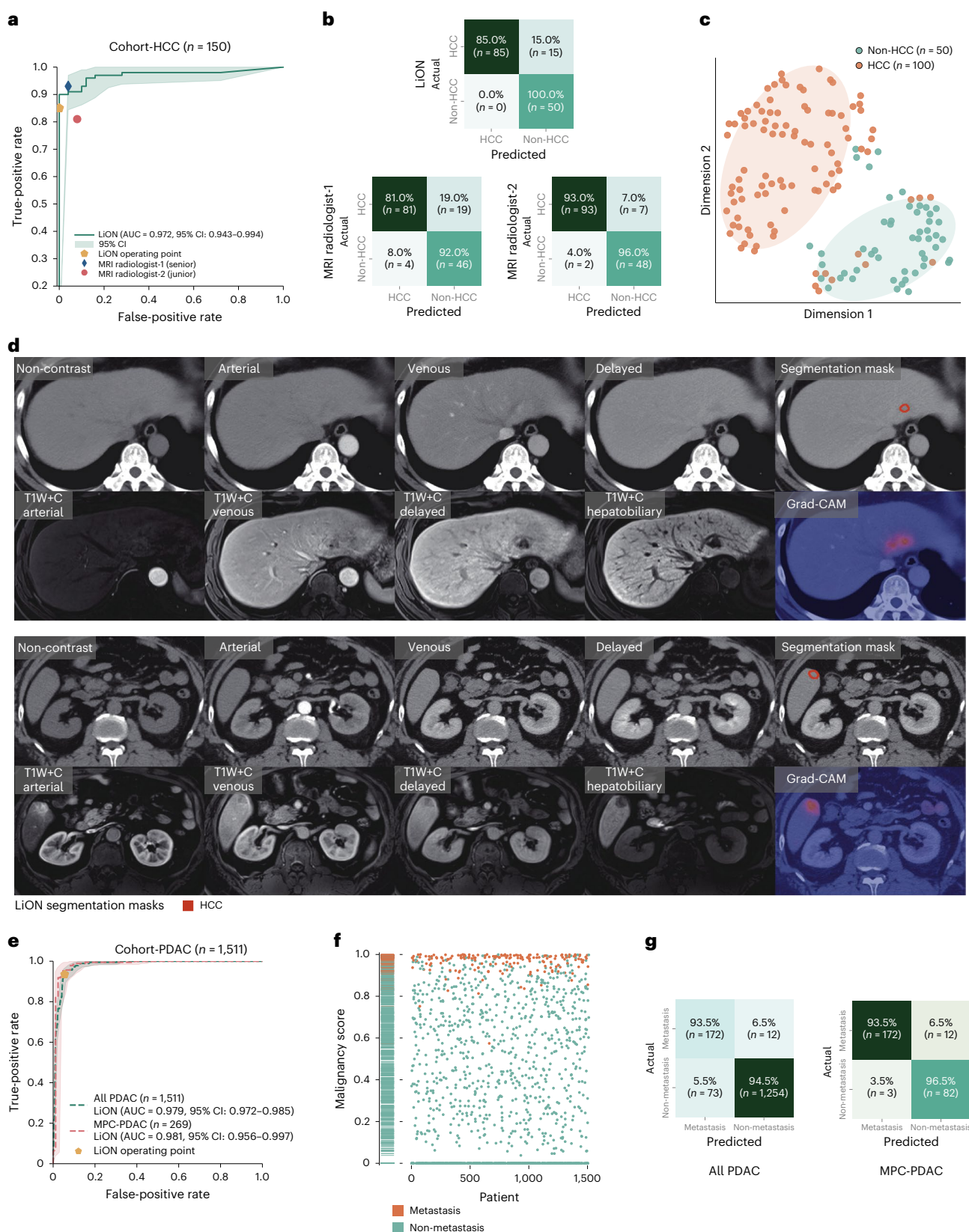


Fig. 3 | Extended validation. a–d, LiON in the diagnosis of early-stage HCC compared to MRI in Cohort-HCC ($n = 150$): comparison of LiON performance with MRI radiologists (**a**); binary confusion matrices for LiON and radiologists using MRI in malignancy diagnosis (**b**); t-SNE visualization of patient-level features for patients with HCC and patients without HCC (**c**); and demonstration of LiON segmentation results on CE-CTs with corresponding MRI images (**d**). **e–g**, LiON performance in the diagnosis of pancreatic metastases in Cohort-PDAC ($n = 1,511$):

receiver operating characteristic curves for LiON (**e**); distribution of LiON scores among patients with PDAC with and without hepatic metastases, with the density distribution shown in the left margin (**f**); and binary confusion matrices for LiON in malignancy diagnosis (**g**). In **a** and **e**, the metrics are reported with 95% CIs from 2,000 bootstrap replications. Error bars represent 95% CIs. T1W+C, contrast-enhanced T1-weighted magnetic resonance imaging.

In the consecutive-patient cohorts (RWR1 and RWR2), LiON confirmed its utility as a real-world diagnostic safety net. At the pre-specified operating point, the system achieved a negative predictive value (NPV) of 99.0% (95% CI: 98.8–99.2%) and a positive predictive value (PPV) of 50.4% (95% CI: 47.6–53.3%). The ability of LiON to exclude liver malignancy was consistent with the observed diagnostic performance, with an AUC of 0.953 (95% CI: 0.945–0.960), a specificity of 95.8% (95% CI: 95.5–96.2%) and a sensitivity of 81.2% (95% CI: 78.3–83.9%). The system maintained high accuracy in challenging subgroups, including patients with cirrhosis (AUC 0.924, 95% CI: 0.901–0.946) and hepatic steatosis (AUC 0.971, 95% CI: 0.952–0.985). Furthermore, LiON showed strong performance across other key capabilities, including eight-class classification (micro-average AUC 0.947, 95% CI: 0.944–0.950), lesion screening (AUC 0.944, 95% CI: 0.941–0.947) and diagnosis of specific major liver malignancies such as HCC (AUC 0.978, 95% CI: 0.935–0.992) (Fig. 4a–e).

Beyond performance evaluation, we tested the utility of LiON as an active screening tool in the targeted high-risk cohorts (Cohort-RWR3 and Cohort-RWR4). Longitudinal follow-up confirmed one malignancy in Cohort-RWR3 and four malignancies in Cohort-RWR4. LiON identified four previously undiagnosed malignancies: one new malignancy was detected in the oncology follow-up cohort (Cohort-RWR3, one of eight predicted; Extended Data Fig. 6a), and three new malignancies were found in the high-risk cirrhotic cohort (Cohort-RWR4, three of 15 predicted; Extended Data Fig. 6b–d). These findings suggest the potential of LiON to enhance surveillance programs by finding occult malignancies.

Single-arm trial. To evaluate the real-world clinical impact of LiON, we conducted a single-arm trial at SH-CMU. Between September and October 2025, we consecutively enrolled adult patients (≥ 18 years) undergoing CE-CT for abdominal assessment with liver coverage. Eligible participants were required to have complete imaging data including, at minimum, non-contrast, arterial and venous phases. We excluded patients with a history of recent major hepatic surgery or trauma (within 30 days), severe image artifacts or inadequate scanning ranges.

The trial was pilot deployed into daily clinical workflow as an internal diagnostic support system. LiON analyzed scans in parallel with standard-of-care radiological interpretation (Extended Data Fig. 1c). Discordance between LiON findings and the primary radiology report triggered the interventional component: discrepant cases were reviewed by a senior radiologist, with unresolved or clinically relevant findings escalated to the MDT for consensus review and clinical action recommendations (Fig. 4h). The discordance was mainly categorized into two types: detection discordance (overlooked lesions) and diagnostic discordance (for example, malignant versus non-malignant and borderline). This workflow ensured that necessary amendments were processed internally before finalizing reports for patient management. The primary outcome was patient-level diagnostic accuracy for malignancy diagnosis, measured by AUC with prespecified success criterion of lower 95% CI bound greater than 0.900. Secondary outcomes included (1) diagnostic performance metrics (sensitivity, specificity, NPV and PPV); (2) lesion detection performance (AUC); and (3) clinical impact, measured by the number of AI-detected missed lesions, report amendments and management changes. Safety outcomes were defined as adverse events, including physical harm from unnecessary procedures, radiation exposure from unwarranted imaging (primarily PET/CT), substantial psychological distress from false-positive findings or any more than 7-day delay in diagnosis or treatment caused by the triggered MDT review.

For the primary outcome, a total of 10,333 eligible participants were enrolled in the full analysis set. LiON demonstrated a patient-level AUC of 0.952 (95% CI: 0.942–0.961) for malignancy diagnosis (Fig. 4f), meeting the predefined success criterion and providing the necessary

context for evaluating subsequent workflow and clinical decision support outcomes. For secondary outcomes, the system achieved a sensitivity of 82.7% (95% CI: 79.3–85.8%), a specificity of 95.3% (95% CI: 94.9–95.7%), PPV of 49.2% (95% CI: 45.8–52.4%) and NPV of 99.0% (95% CI: 98.8–99.2%) for malignancy diagnosis and an AUC of 0.961 (95% CI: 0.958–0.965) for lesion screening. For clinical impact, 407 of 10,333 (3.9%) cases were flagged for diagnostic discordance and 152 (1.5%) for detection discordance between LiON and initial radiological reports during the intervention phase. After senior radiologist and MDT adjudication, the AI–human collaboration resulted in the following clinical impacts. (1) The system identified 51 confirmed, previously overlooked lesions, including 15 small metastases (1.45 additional malignancies per 1,000 CE-CT examinations). (2) After clinical relevance assessment, 37 actionable cases prompted internal formal addenda to the radiology reports in the electronic medical records before final clinical decision-making. (3) Among the 15 patients with newly discovered metastases, immediate therapeutic interventions were initiated for four patients due to disease upstaging (0.04%): one underwent surgical resection for colorectal metastasis, and three (with primary breast, bladder or colon malignancies) had treatment regimens modified (representative cases are shown in Figs. 4h and 5a). (4) Additionally, among 22 escalated MDT reviews (0.21%), 14 patients were recalled for evaluation, leading to five new confirmed malignancy diagnoses (0.05%; a representative case is shown Fig. 5b). No AI-attributable adverse events occurred under mandatory human review. All discordant findings underwent mandatory review by a senior radiologist, and unresolved or clinically relevant cases were escalated to MDT review before clinical action, safeguarding against unnecessary procedures triggered by false-positive AI findings.

Notably, discordant cases frequently involved lesions with ambiguous imaging characteristics. We, therefore, performed a subgroup analysis of diagnostically uncertain cases, separating them into two categories according to the degree of uncertainty in radiologists' assessment. Borderline cases, denoted by open circles ($\circ$), were lesions with atypical or incomplete imaging features for which radiologists nevertheless provided a preliminary benign or malignant impression. Highly ambiguous or indeterminate cases, denoted by filled circles ($\bullet$), were lesions considered insufficient for a definitive diagnosis, often requiring further examination or follow-up. These scenarios are clinically challenging because such lesions do not clearly conform to typical benign or malignant imaging patterns. Accordingly, clinical management may vary, with indeterminate lesions such as LI-RADS 3/4 observations often requiring short-interval imaging follow-up or additional diagnostic workup, including biopsy¹².

First, we analyzed the borderline cases ($\circ$), including 145 malignant and 51 benign lesions according to the reference standard. In this subgroup, LiON and radiologists showed complementary performance patterns (Fig. 4g). Radiologists achieved significantly higher sensitivity ($P < 0.01$), consistent with a conservative strategy to avoid missed malignancies, whereas LiON showed significantly higher specificity ($P < 0.01$), suggesting added value in reducing false-positive malignancy calls. This complementarity identified a higher-confidence subset of concordant AI–human diagnoses, with higher agreement with the reference standard ($\kappa = 0.693$, 95% CI: 0.538–0.832) compared to radiologists alone ($\kappa = 0.476$, 95% CI: 0.334–0.620). When both LiON and radiologists agreed on a non-malignant diagnosis, the NPV reached 100% (20/20).

Second, we analyzed the highly ambiguous or indeterminate cases ($\bullet$), including 149 benign and 65 malignant lesions according to the reference standard. In this subgroup, radiologists considered the imaging findings too uncertain for a definitive diagnosis and often recommended further examination, with representative examples resolved by follow-up imaging (Extended Data Fig. 7). In this challenging group, LiON maintained high specificity (95.3%, 95% CI: 91.6–98.1%), with sensitivity of 84.6% (95% CI: 75.3–92.7%), supporting its value in identifying non-malignant cases among diagnostically indeterminate lesions.

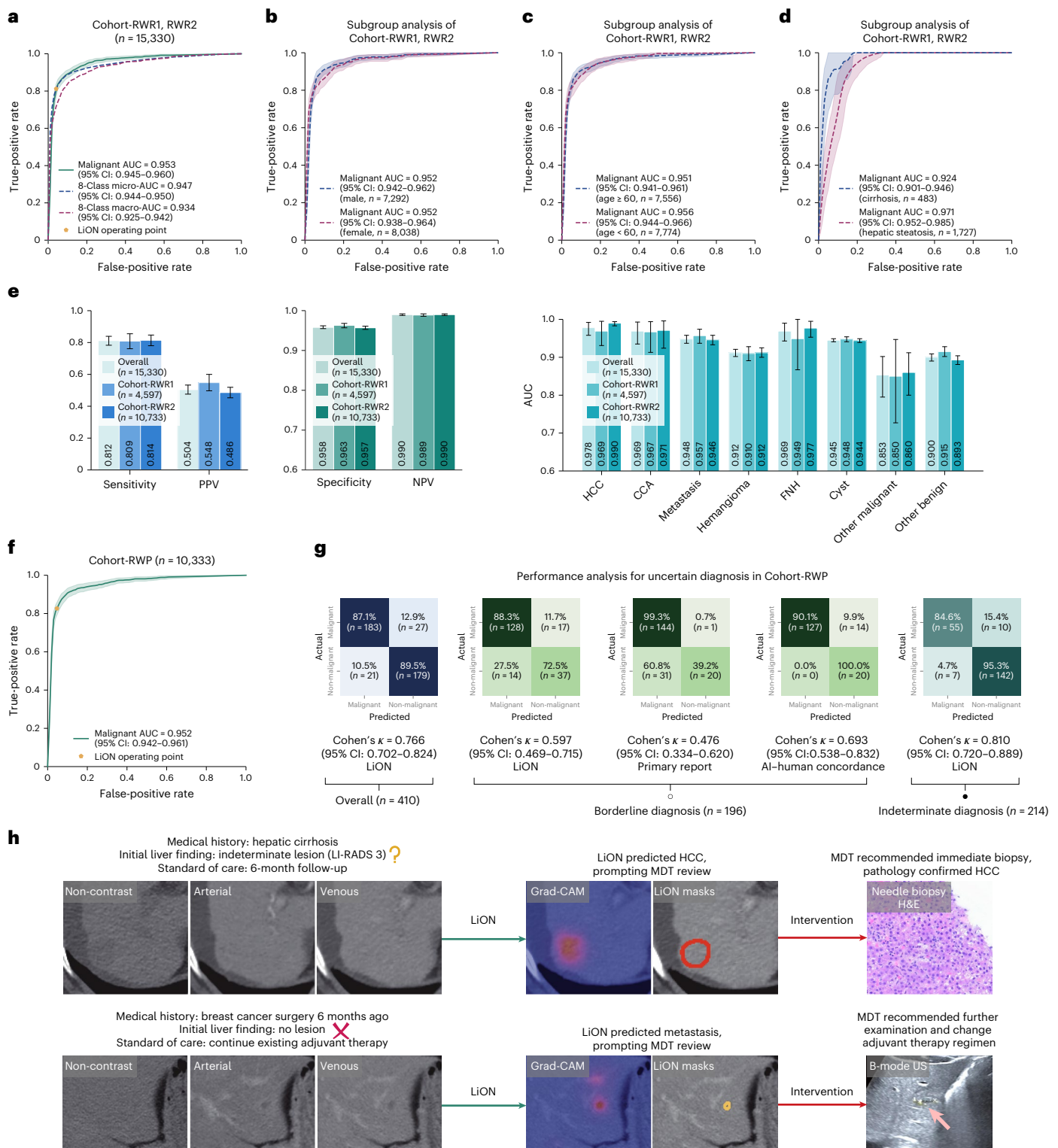


Fig. 4 | Real-world validation. **a–d**, Performance in the retrospective cohorts (n = 15,330): receiver operating characteristic curves for the overall cohort (**a**) and subgroup analyses stratified by sex (**b**), age (**c**) and complex liver background (**d**). **e**, Bar plots for overall diagnostic metrics (sensitivity, PPV, specificity, NPV; left) and the AUC for specific lesion types (right). **f,g**, Performance in the prospective study: receiver operating characteristic curve for malignancy diagnosis in Cohort-RWP (n = 10,333) (**f**); binary confusion matrices for LiON in malignancy diagnosis on diagnostically challenging cases (**g**). Cases were

selected based on the initial radiological report, which either indicated diagnostic uncertainty while still providing a preliminary impression of malignancy or benignity (•) or recommended further examination for a definitive diagnosis (•). For all selected cases, the reference standard was the final diagnosis determined by subsequent examination or MDT consensus, blinded to the results of LiON. **h**, Examples of AI-human collaboration leading to altered clinical pathways. In **a–f**, the metrics are reported with 95% CIs from 2,000 bootstrap replications. Error bars represent 95% CIs. H&E, hematoxylin and eosin.

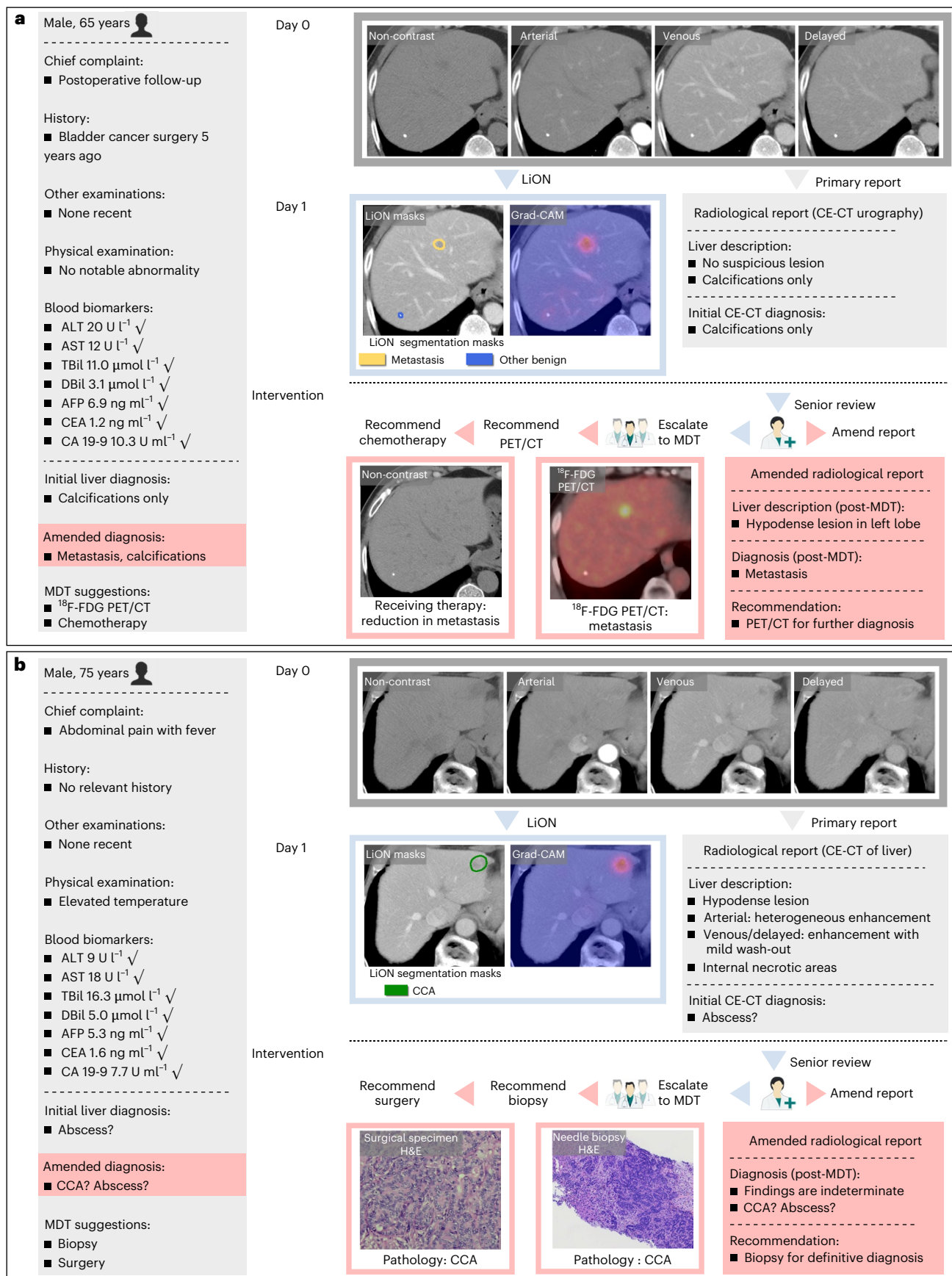


Fig. 5 | Representative intervention in Cohort-RWP. a, A 65-year-old patient in whom a previously overlooked metastasis was detected, leading to modified treatment decisions by the MDT. **b**, A 75-year-old patient in whom an initially uncertain diagnosis was escalated to MDT review, leading to an immediate

biopsy recommendation, which confirmed malignancy, followed by surgical intervention. ¹⁸F-FDG PET/CT, ¹⁸F-fluorodeoxyglucose positron emission tomography/computed tomography; ALT, alanine aminotransferase; AST, aspartate aminotransferase; DBil, direct bilirubin; TBil, total bilirubin.

These findings suggest the potential for a clearer clinical pathway. Concordance between LiON and radiologists provides high-confidence confirmation (for example, expediting intervention for malignant cases or avoiding unnecessary follow-ups for benign ones). Conversely, a discordant assessment can flag complex cases for MDT review, providing diagnostic scrutiny exactly where it is needed.

Finally, we characterized the 51 overlooked lesions by their lesion-level features (Extended Data Fig. 8). These lesions were generally small (diameter of 1.00 ± 0.42 cm) and showed poor enhancement, and 33.3% were located in anatomically challenging liver segments (for example, segment VI). Specifically, most oversights occurred during non-liver-specific examinations (for example, general abdomen and urography), where attention may be primarily directed toward the indication-specific target. These findings highlight the potential value of LiON as a diagnostic support for liver malignancy in high-volume, real-world radiology workflows.

Discussion

We have demonstrated that LiON, an AI system for diagnosing overall patient-level liver malignancy and segmenting eight liver lesion subtypes on CE-CT, can be integrated into clinical practice. LiON was effective across CE-CT examinations with liver coverage, despite variation in scanning protocols and clinical indications. Our study confirmed its robust diagnostic performance across multicenter and real-world populations, particularly for major malignancies (for example, HCC and metastasis). In particular, LiON functioned as a workflow-compatible diagnostic support: a collaborative AI-human framework that reduces overlooked malignancies while serving as a trusted second opinion for indeterminate cases. The system demonstrated potential to influence clinical decision-making.

Recent studies have established multicenter retrospective validation for AI-assisted focal liver lesion diagnosis and highlighted the potential of AI for triage and diagnostic support^{22,23,25}. LiON extends this foundation through both architectural and methodological designs. Architecturally, unlike previous approaches restricted to fixed CT phase combinations^{22,24}, LiON supports flexible-phase input, and delayed-phase information significantly improved diagnostic accuracy for lesion subtypes (for example, metastasis and hemangioma). Furthermore, our end-to-end architecture improved processing speed and reduced error propagation compared to previous multistage architectures. Methodologically, LiON employs a finer-grained lesion taxonomy addressing the full spectrum of FLL subtypes rather than grouping rare subtypes. Notably, LiON was validated through both retrospective validation and prospective deployment. In the real-world cohort, LiON showed clinically useful operating characteristics, as it achieved a PPV of 50.4% while maintaining a high NPV of 99.0%. LiON did not rely on explicit center-specific adaptation to handle data heterogeneity. Instead, the diversity of the training data, a standardized preprocessing pipeline and the phase-flexible model design facilitated generalization. Consequently, our prospective study suggests a framework that moves beyond the ‘rule-out’ triage of low-risk patients, demonstrating that AI-human collaboration also enabled escalation of suspicious cases and clinical action.

The robustness of LiON was evaluated in several high-stakes clinical scenarios. In the stress-test cohort for early-stage HCC, where timely diagnosis critically affects treatment eligibility and survival⁵, LiON achieved diagnostic performance approaching radiologists’ interpretation of CE-MRI (Gd-EOB-DTPA) using CE-CT inputs alone. This finding has practical relevance, as early-stage HCC (BCLC 0-A) is amenable to potentially curative treatment and is associated with substantially improved 5-year survival, whereas advanced-stage disease is associated with markedly worse outcomes and predominantly palliative treatment options⁵. As CE-MRI remains the preferred imaging modality for early-stage HCC and offers higher diagnostic sensitivity²⁵,

this comparison served as a clinical benchmark rather than a direct equivalence test between CE-CT and CE-MRI. At the same time, CE-CT is frequently performed in routine abdominal and oncologic workflows, including non-hepatic indications, enabling opportunistic liver assessment, whereas CE-MRI generally requires a dedicated protocol and may be limited by availability, cost, contraindications or workflow constraints. In this context, LiON may help narrow the diagnostic gap between routine CE-CT and MRI assessment by improving CT-based risk stratification, particularly in settings with limited MRI accessibility. Metastases are more common than primary liver malignancies², a pattern also observed in our real-world cohorts, where metastases accounted for 75–85% of liver malignancies. Accordingly, LiON was developed and validated to detect hepatic metastases from diverse primary origins. We further evaluated Cohort-PDAC as a focused stress test, because hepatic metastasis is a key determinant of PDAC staging and treatment eligibility, with the liver representing the most frequent metastatic site³⁶. This setting is clinically relevant, as CE-CT is the recommended modality for PDAC resectability assessment and preoperative evaluation of hepatic involvement³⁴, yet its diagnostic performance remains moderate³⁵. In this scenario, LiON achieved 93.5% sensitivity and 94.5% specificity, suggesting potential to complement radiologist interpretation. Nevertheless, our reference standards relied on histopathological confirmation or, when biopsy was contraindicated in patients with MPC³⁷, MDT review, which may have favored cases with more distinctive imaging features.

A major clinical challenge, acknowledged by international guidelines, is the management of indeterminate FLLs. Current guidelines recommend risk stratification and MDT evaluation for lesions showing high-risk but non-diagnostic features^{7,10,12}. The management of these indeterminate lesions varies by guideline. For instance, the American Association for the Study of Liver Diseases (AASLD) recommends MDT discussion for LI-RADS 4 observations¹², whereas the American College of Gastroenterology (ACG) and the European Association for the Study of the Liver (EASL) advocate biopsy for atypical lesions in patients with cirrhosis^{8,10}. In practice, this variance creates a clinical dilemma between invasive biopsy and prolonged surveillance¹², as managing these borderline lesions is difficult, and the optimal path forward is often ambiguous. Our prospective study demonstrated that LiON could trigger MDT review in cases requiring escalated diagnosis (including those with discordant readings, borderline features or potential malignancy suggested by either AI or radiologists), ensuring timely expert evaluation for challenging cases that might otherwise remain in prolonged surveillance (Fig. 4g). Furthermore, in the prospective study, among the malignancies identified by LiON that were initially overlooked in clinical practice, all were metastases, suggesting that secondary hepatic malignancies remain particularly challenging to detect. The radiological differentiation of early-stage metastases from benign lesions is challenging¹³, particularly as their imaging characteristics can vary widely with different primary origins. In our prospective study, the initially overlooked metastases were small (diameter ≤ 2.25 cm), yet their clinical implications are substantial. Even small hepatic metastases indicate primary malignancy progression and require major therapeutic strategy modifications, with important implications for treatment planning and prognosis. For example, colorectal cancer is a leading primary source of hepatic metastases, and untreated hepatic metastases are associated with poor prognosis, whereas appropriate treatment could prolong survival^{38,39}. From a health-economic perspective, earlier escalation to curative-intent treatment may reduce downstream medical expenditure compared to delayed diagnosis requiring repeated locoregional and targeted therapy⁴⁰. Finally, subgroup analysis showed that LiON maintained consistent performance across clinically relevant subgroups (for example, viral hepatitis and metabolic dysfunction-associated steatotic liver disease (MASLD)), supporting its potential utility in surveillance and screening workflows

for high-risk populations, particularly in settings where first-line screening modalities (for example, ultrasound) may be limited by coarse hepatic echotexture or severe acoustic attenuation.

Despite the robust performance of LiON, several limitations should be discussed transparently. First, our validation was conducted on cohorts from a Chinese population; therefore, its generalizability to other ethnic groups and diverse international cohorts requires further validation. In addition, as our study was based on guideline-consistent diagnostic multiphasic CE-CT protocols^{7,29,41,42}, broader real-world validation remains warranted in venous-phase-only examinations (for example, oncology follow-up and non-liver-specific abdominal examinations). Second, in the real-world deployment, we observed lower specificity in patients with cirrhosis and diagnostic challenges in distinguishing malignancies from rare non-malignant conditions (for example, atypical nodular focal fat and hepatic arteriovenous fistula). Similarly, detection sensitivity for certain rare benign lesions (for example, hepatic adenoma and regenerative/dysplastic nodules) was relatively lower. Third, to ensure real-time processing, the prospective study was performed without the integration of clinical information, a component that our analysis showed to be beneficial. Clinical information was often unavailable for real-time retrieval due to asynchronous data workflows or incomplete testing in routine practice. Consequently, the post hoc analysis revealed that the image-only approach introduced mild conservative bias in diagnostic performance (Supplementary Fig. 20). Future studies will consider an asynchronous multimodal inference pipeline to incorporate clinical information without compromising real-time performance. Fourth, because threshold-dependent performance may vary under real-world distribution shifts^{43,44}, prospective calibration monitoring during clinical deployment is warranted. Finally, our study is limited by its single-arm design and a relatively short follow-up period. As a result, comparative workflow timeliness and clinical outcomes (for example, time-to-treatment) could not be comprehensively evaluated. Although our study demonstrated that AI–human collaboration can alter clinical management, a randomized controlled comparison is necessary to quantify its impact against the standard of care. We are working with multicenter clinical partners to design a prospective randomized controlled trial across a broader range of medical centers for comprehensive safety and cost-effectiveness evaluation.

Future studies should assess generalizability across broader clinical settings, strengthen performance for rare and challenging lesion subtypes, extend the approach to longitudinal applications such as post-treatment surveillance and recurrence monitoring after locoregional or systemic therapy and adapt the system to a broader range of CT protocols and available phase combinations. Rather than replacing clinicians, LiON may be most valuable within a collaborative workflow that helps reduce overlooked malignancies, supports the triage of indeterminate findings for further evaluation and preserves clinician oversight. Together, these findings support the clinical translation of AI-assisted liver malignancy diagnosis and provide a practical basis for integrating AI into existing diagnostic workflows, with a focus on clinically consequential failure modes.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at <https://doi.org/10.1038/s41591-026-04589-y>.

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Methods

Ethics approval

All research activities adhered to the principles of the Declaration of Helsinki and institutional guidelines. This study was conducted under institutional review board approval from all participating centers. The prospective component was approved by the institutional review board of SH-CMU and registered at ClinicalTrials.gov (NCT07153783), and it adhered to CONSORT-AI guidelines⁴⁵. The ethics committees waived informed consent requirements for retrospectively collected CT images and clinical data. Written informed consent was obtained from patients whose clinical information, CE-CT examinations and follow-up records were prospectively collected.

Dataset description

This study utilized a multicenter dataset to develop and validate LiON (Extended Data Fig. 2) for FLL diagnosis using CE-CT. In total, 39,027 patients with 139,801 CT scans were included. Detailed patient characteristics and the nomenclature of each cohort are provided in Table 1, Extended Data Table 1 and Supplementary Table 1, respectively. To ensure broad geographical representation and minimize regional bias, we collected data from six regions across China (Northeast, East, South, North, Central and Southwest) through a consortium of seven major medical centers (Extended Data Fig. 2). The external centers included two major oncology specialist hospitals, Hubei Cancer Hospital (HCH) and Sun Yat-sen University Cancer Center (SYSUCC), along with three general hospitals: The Second Affiliated Hospital of Baotou Medical College (SAH-BMC), Yantai Yuhuangding Hospital (YYH) and Chengdu Sixth People's Hospital (CSPH). The PDAC data for hepatic metastasis diagnosis were obtained from Shanghai Institution of Pancreatic Diseases (SIPD).

The training cohorts, comprising 6,443 patients from two general medical centers (SH-CMU and FAH-ZU), were used for model development. The training cohorts included patients over 18 years of age who had undergone complete multiphase CE-CT examinations, with malignancy confirmed by pathological analysis and benign lesions confirmed either by pathology or by a minimum 1-year imaging follow-up. No patients or scans overlapped between the training cohort and any validation cohort. Of these, the primary training dataset included 3,669 malignant cases, 2,022 benign cases and 752 no-lesion cases. The dataset encompassed diverse lesion pathologies. Common malignancy diagnoses included HCC, CCA and metastasis, and benign lesions primarily consisted of hemangioma, FNH and cysts. Lesion volumes differed between centers, with SH-CMU showing larger mean volumes ($48.87 \pm 192.61 \text{ cm}^3$ versus $25.68 \pm 102.11 \text{ cm}^3$). To address the challenge of varying scanning protocols across different centers, we collected both three-phase (non-contrast, arterial, venous, without delayed phase) and four-phase (with delayed phase) CE-CT scans. The training set included 2,567 three-phase scans and 3,876 four-phase scans (average 3.60 scans per patient), with HCH predominantly contributing three-phase scans because of variations in institutional examination protocols.

For comprehensive multicenter validation, two major cohorts were included: a held-out internal validation set (Cohort-IN, $n = 300$) of randomly selected cases from the internal centers (SH-CMU ($n = 124$) and FAH-ZU ($n = 176$)) and an external test set (Cohort-EX, $n = 4,071$) collected from five external centers (HCH ($n = 2,401$), SAH-BMC ($n = 740$), SYSUCC ($n = 654$), YYH ($n = 219$) and CSPH ($n = 57$)). To assess performance in clinically challenging scenarios, we included Cohort-HCC ($n = 150$), a small-lesion cohort comprising patients from SH-CMU with lesions $\leq 3 \text{ cm}$ in diameter, and Cohort-PDAC ($n = 1,511$) from SIPD. Cohort-PDAC comprised 546 patients with resectable pancreatic cancer, 123 with borderline resectable pancreatic cancer, 573 with locally advanced pancreatic cancer and 269 with MPC, including 184 with hepatic metastases. Hepatic metastasis was confirmed through histopathological biopsy or, when biopsy was contraindicated

or unfeasible in advanced PDAC, through comprehensive MDT evaluation of clinical evidence.

For real-world performance assessment, we established testing cohorts from internal centers (FAH-ZU and SH-CMU), encompassing 26,552 patients over 18 years of age who underwent CE-CT with liver coverage for various clinical indications, representing diverse clinical scenarios (outpatient, emergency and inpatient). The inclusion criteria encompassed all CE-CT scans involving the liver, including abdominal scans with different anatomical coverage (for example, pelvis and upper/lower abdomen), and organ-specific protocols (for example, pancreatic and urographic scans). Patients with major liver surgery and acute injury were excluded. The detailed process for establishing reference standards is described in Extended Data Fig. 5.

LiON algorithm

The LiON architecture consists of an image encoder, a backbone network and dual-task modules for lesion segmentation/detection and malignancy scoring (Extended Data Fig. 3). The image encoder is designed to accommodate variable input phases, enabling flexible processing. Input phases are automatically identified without requiring manual labels. For accurate diagnosis, we set the minimum required inputs as non-contrast, arterial and venous phase images, and LiON was trained end-to-end using this encoder. The backbone leverages feature pyramid networks⁴⁶ for more efficient multiscale processing. For lesion-level analysis, the segmentation head serves as the detection module. In parallel, the classification branch analyzes the entire liver volume to generate patient-level predictions for malignancy assessment and lesion subtype classification. The overall architecture builds upon our previous studies^{47,48}.

Image encoder. In clinical practice, radiologists characterize FLLs by evaluating key enhancement patterns and their changes across multiple imaging phases. Inspired by this diagnostic process, we developed an iterative fusion strategy to integrate complementary information from different phases of CE-CT scans. Standard CE-CT protocols typically include non-contrast, arterial and venous phases; delayed-phase imaging can provide additional information for lesion subtype diagnosis.

The image encoder uses an iterative fusion mechanism to extract and enhance pixel-level changes across available phases⁴⁸. To further emphasize inter-phase differences, each input feature is disentangled into similar and dissimilar components⁴⁹. Similar components are directly aggregated, whereas dissimilar components are processed through a cross-attention mechanism^{50,51}. This design enhances phase-specific feature differentiation while making maximal use of the available imaging phases. The resulting multiphase representation captures both shared and phase-distinctive features, supporting comprehensive lesion characterization.

Backbone. LiON's backbone architecture builds upon the encoder design, with modifications to improve computational efficiency for high-resolution three-dimensional medical volume analysis. The encoder consists of six levels of three-dimensional convolutional layers, each followed by pooling operations. This hierarchical structure progressively reduces spatial resolution while increasing feature depth, enabling multiscale contextual feature extraction for liver anatomy and lesion heterogeneity across different imaging phases.

To address the memory demands of processing high-resolution three-dimensional volumes, LiON departs from the traditional symmetric nnU-Net decoder. Instead, it adopts a Feature Pyramid Network (FPN) decoder⁴⁶, which generates multiscale feature maps (P1–P6). This design provides a more memory-efficient mechanism for aggregating multi-resolution information, facilitating the processing of larger volumes. The FPN constructs a hierarchical feature pyramid by integrating semantic information from deeper encoder-derived

features (C3–C6) with fine-grained spatial details from shallower features (C1–C2) through lateral connections.

This multiscale representation provides a shared feature foundation for downstream lesion segmentation and classification. In particular, it preserves global liver context while retaining local texture and boundary information, which are both important for multiphase liver lesion characterization.

Instance-level segmentator. Our approach to instance-level segmentation builds upon mask transformer architectures⁴⁷, predicting both a class label and a binary mask for each identified object instance. The self-attention component enables each lesion instance to learn from other lesions, while the cross-attention mechanism captures contextual relationships between lesions and surrounding organs (for example, liver), thereby enhancing the representation of fine-grained semantic relationships and spatial dependencies.

The segmentation head comprises a pixel encoder and decoder, which extract a high-resolution pixel embedding tensor $\mathbf{P} \in \mathbb{R}^{M_{\text{seg}} \times D \times H \times W}$ from the input image, where M_{seg} is the embedding dimension and $D \times H \times W$ denotes the three-dimensional image dimensions. A set of Q_{seg} learnable query embeddings, $\{\mathbf{q}_i^{\text{seg}} \in \mathbb{R}^{M_{\text{seg}}}\}_{i=1}^{Q_{\text{seg}}}$, is initialized as object queries. These queries are then processed by a stacked transformer decoder. As depicted in Extended Data Fig. 3d, the decoder integrates multiscale features (P2, P3, P4) extracted from the FPN. Through sequential self-attention and cross-attention operations, the queries are iteratively refined to aggregate information from different image resolutions and capture object-specific context. Each refined query $\mathbf{q}_i^{\text{seg}}$ thus encodes the representation of a candidate object instance.

This object-specific information is used to predict the class probability $c_i \in \mathbb{R}^{C_{\text{seg}}+1}$ for object i , where C_{seg} is the number of object classes, with an additional no-object class for queries not corresponding to any ground truth object. The binary mask $\mathbf{m}_i \in \mathbb{R}^{D \times H \times W}$ for object i is generated through matrix multiplication of the refined query $\mathbf{q}_i^{\text{seg}}$ with the pixel embedding tensor $\mathbf{P}$. For inference, class predictions and binary masks from all queries are merged via matrix multiplication, yielding the final instance-level semantic segmentation result $\hat{\mathbf{Y}} \in \mathbb{R}^{C_{\text{seg}} \times D \times H \times W}$.

The overall loss function for the instance-level segmentator $\mathcal{L}_{\text{seg}}$ is designed to simultaneously optimize instance classification, mask prediction and instance discrimination. This loss is applied across multiple decoder layers (deep supervision) and utilizes a bipartite matching strategy during training. For a given image, let Q_{seg} denote the total number of object queries and N the number of ground truth objects. During training, the Hungarian algorithm is used to find the optimal assignment σ that minimizes the matching cost between the Q_{seg} predicted queries and the N ground truth objects. The set of matched query–ground truth pairs is denoted as $\{(\hat{y}_{\sigma(i)}, y_i)\}_{i=1}^N$, where $\hat{y}_{\sigma(i)}$ represents the prediction for the matched query and y_i is the corresponding ground truth. Unmatched queries are assigned to a no-object class.

Patient-level classifier. LiON incorporates a parallel patient-level classification head, complementing the instance-level segmentation, which provides diagnostic predictions for the entire input, leveraging multiscale features and a transformer-based architecture. To further enhance patient-level diagnosis, we implement a two-stage machine learning approach using logistic regression for score-based fusion, which can optionally integrate clinical data.

As illustrated in Extended Data Fig. 3e, the classification head processes multiscale image features (P5, P3, C5, C3), representing different levels of feature abstraction. A set of Q_{cls} learnable query embeddings, $\{\mathbf{q}_i^{\text{cls}} \in \mathbb{R}^{M_{\text{cls}}}\}_{i=1}^{Q_{\text{cls}}}$, is initialized, distinct from those in the segmentation head. These queries are processed by a stacked transformer decoder, in which self-attention models inter-query

relationships and cross-attention integrates information from the multiscale image features. Through iterative refinement, the queries aggregate global context relevant to patient-level diagnosis. The refined query embeddings are then pooled into a compact representation and passed to a lightweight classifier to generate image-level probabilities for malignancy assessment and lesion subtype classification.

When clinical data are available, we apply a two-stage logistic regression framework to integrate clinical information with image-derived probability scores. In Stage I, a clinical-only logistic regression model is trained using clinical information (that is, demography, blood biomarkers and medical history) without incorporating any image-based probability scores. The full clinical features are provided in Supplementary Table 4. The output probabilities from this clinical model are then combined with the image-derived probability scores in Stage II, where the second logistic regression model further refines the diagnostic predictions. This second stage focuses on eight diagnostic categories (that is, malignancy, HCC, CCA, metastasis, hemangioma, cyst, FNH and others). This hierarchical design enables clinical information to first provide an independent diagnostic estimate, which is subsequently integrated with image-based predictions for more accurate diagnosis.

Annotations

To manage the substantial segmentation workload while maintaining annotation quality, we implemented a paired annotation strategy. The expert panel, comprising two senior diagnostic radiologists and two senior hepatic surgeons, was organized into two radiologist–surgeon pairs. Each pair was assigned a portion of the dataset. Within each pair, one expert performed initial three-dimensional lesion delineation on CE-CT venous phase images while their counterpart reviewed and refined these contours. Segmentations were finalized only upon reaching consensus within each pair. This collaborative approach integrated radiological and surgical expertise to establish segmentation ground truth.

Lesion-level diagnosis ground truth was determined through histopathological reports and complementary imaging findings (MRI and PET). Visible lesions with clear segmentation but indeterminate diagnosis were retained in the dataset with an ‘unknown’ label. Additionally, we annotated anatomical context, including five perihepatic organs (gallbladder, spleen, stomach, pancreas and right kidney) and major vessels (the inferior vena cava, portal veins and other intrahepatic vessels). Initial masks for the perihepatic organs and inferior vena cava were generated using pretrained anatomical segmentation models⁵². For hepatic vasculature, a representative subset of cases was manually annotated and used to train an in-house vessel segmentation model using nnU-Net¹⁶, which then produced initial vessel masks for the remaining cases. All automatically or semiautomatically generated masks were subsequently reviewed and manually refined by the assigned radiologist–surgeon pairs. Final anatomical segmentations were accepted only after consensus within each pair.

LiON pilot deployment

For the single-arm trial, the detailed protocol is provided in Supplementary Information. The study was conducted in SH-CMU from 1 September to 29 October 2025, with the first participant enrolled on 1 September 2025 and the last participant enrolled on 29 October 2025, when target enrollment was reached.

The sample size was calculated to provide 90% statistical power ($1 - \beta$) to test the one-sided superiority hypothesis (H_1 : AUC > 0.900) against a benchmark of 0.900, at a significance level of $\alpha = 0.025$. Assuming a malignancy prevalence of $\approx 5.0\%$, an expected AUC of 0.950 based on the real-world retrospective study and a 20% rate of data loss or attrition, a target enrollment of approximately 3,951 participants was required (using PASS 2022; NCSS, LLC). We consecutively enrolled all

eligible patients during the 2-month study period, anticipating approximately 10,000 participants based on institutional patient volume. This larger enrollment was chosen to improve precision in estimating the primary AUC and to support exploratory subgroup analyses. Finally, 10,333 participants were enrolled in the full analysis set.

To establish ground truth diagnoses in this real-world setting, a composite reference standard was implemented. Each participant was assigned a patient-level binary diagnostic label (malignant versus non-malignant) based on a hierarchy of evidence: (1) histopathology (biopsy or resection); (2) expert panel adjudication, conducted as a de novo review, blinded to both AI output and the initial radiological conclusion, based on comprehensive review of follow-up imaging (for example, MRI and PET/CT) and clinical records; and (3) MDT consensus for equivocal cases. To minimize bias, radiologists randomly reviewed 1% of cases that had been initially classified as non-malignant.

LiON was developed as an automated pipeline for processing CE-CT images and integrated directly with the Picture Archiving and Communication System (PACS) at SH-CMU. To ensure consistent performance, the LiON model (version 2025.02) was frozen prior to enrollment of the first participant and remained unmodified throughout the study. For data acquisition, the system automatically retrieved standard Digital Imaging and Communications in Medicine (DICOM) series from commercial multi-detector CT scanners via the daily PACS workflow.

Image quality control was implemented through a sequential filtering process. First, automated signal-to-noise ratio calculation excluded scans with excessive noise artifacts. Second, valid scans entered the automated technical workflow using established methods for phase classification⁵³ and image registration⁵⁴. To maintain spatial consistency and enable real-time processing, all contrast-enhanced phases were aligned to the non-contrast phase as the reference coordinate system (Supplementary Fig. 5a). The complete workflow comprised (1) contrast phase classification, (2) spatial registration and (3) diagnostic inference by LiON. Cases experiencing technical failures at any pipeline stage (phase classification, registration or inference errors) were excluded from the analysis set. The entire interventional process, from AI analysis to senior radiologist adjudication and potential report amendment, functioned as an internal quality control loop within the radiology department, ensuring that only finalized and verified reports were released to clinical teams and patients.

AI system explainability

The LiON framework operates as an end-to-end system, simultaneously generating instance-level segmentations and overall malignancy predictions. We implement a multifaceted approach to explain these outputs. For instance, segmentations: interpretability is direct through mask overlays on CT images, enabling clinical validation of anatomical boundaries. To elucidate the malignancy classification process, we implement multiple visualization approaches. Specifically, we utilize gradient-weighted class activation mapping (Grad-CAM)⁵⁵ to generate activation heatmaps from the penultimate layer, highlighting the liver regions that most contribute to malignancy classification. The visualization demonstrates that the model's attention primarily focuses on three hemodynamic regions: the focal lesion, perilesional parenchyma and regions of altered hepatic perfusion dynamics. Additionally, to analyze the model's learned representations for patient-wise classification, we extract features from the penultimate layer and apply t-distributed stochastic neighbor embedding (t-SNE)⁵⁶ dimensionality reduction (Fig. 1), visualizing how patients are clustered based on their imaging characteristics.

Computational hardware and software

Experiments were conducted using Python 3.11.5 with PyTorch 2.1.0+cu118, CUDA 12.2 and other libraries, including nnU-Net v1, SciPy 1.11.4, Torchvision 0.16.0+cu118, Scikit-learn 1.3.2, Pandas 2.1.4, NumPy

1.26.2, Scikit-image 0.25.2, SimpleITK 2.4.1, MedPy 0.5.2, Nibabel 5.3.2 and Matplotlib 3.8.2. The hardware configuration consists of four NVIDIA A800-SXM4-80GB GPUs, 256 CPU cores across four NUMA nodes and 512 GB RAM. LiON comprises 35.3 million parameters with a peak memory consumption of 6.07 GB during inference (input shape: $4 \times 35 \times 320 \times 280$), requiring approximately 5 seconds per case for analysis. The complete pipeline, including phase classification, registration and five-fold inference, takes approximately 35 seconds per case (Supplementary Fig. 6).

Statistical analysis

Data analysis was conducted using the Python programming language (version 3.11.5) with NumPy (1.26.2), SciPy (1.11.4) and Scikit-learn (1.3.2) packages. Classification performance was evaluated using the AUC, sensitivity, specificity, PPV, NPV and balanced accuracy. CIs were calculated based on 2,000 bootstrap replications of the corresponding data. Sensitivity and specificity were compared using permutation tests with 10,000 permutations. The threshold for statistical significance was $P < 0.05$.

For the single-arm trial, the primary endpoint was the AUC of the LiON system for malignancy diagnosis. The 95% CI for the AUC was calculated using 2,000 bootstrap replications of the corresponding data. The success criterion was met if the lower bound of the 95% CI was greater than 0.900. Secondary performance metrics included sensitivity, specificity, PPV and NPV. CIs were calculated based on 2,000 bootstrap replications of the corresponding data. Inter-observer agreement was evaluated using Cohen's κ . All secondary hypothesis tests were two-sided with a significance threshold of $P < 0.05$. Prespecified subgroup analyses were performed on lesions with indeterminate or borderline diagnoses to evaluate performance and agreement in difficult cases. Missing data were handled according to the complete-case principle for the primary analysis.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

Due to ethical restrictions protecting patient confidentiality, the datasets analyzed in this study are not publicly available. Qualified researchers may request access to anonymized individual-level data and related clinical documentation by contacting corresponding authors (Qi Zhang, Y. Hou, T.L. and Y.S.). Requests will be evaluated by the relevant institutional data governance committees based on compliance with institutional and regulatory requirements. Access will require approval from the relevant institutional review board where applicable and completion of a data use agreement. Requests will be processed within a maximum of 6 weeks.

Code availability

The full source code for LiON cannot be publicly released due to dependencies on proprietary internal infrastructure (for example, phase classification, image registration and rendering) and pending patent protection (CN116993663A). All experiments and implementation details are described in Methods and Supplementary Information to support replication using non-proprietary libraries. The key algorithmic components and inference logic for FLL detection and diagnosis are available in the open-source repository (<https://github.com/alibaba-damo-academy/pixel-lesion-patient-network>). The framework was implemented in PyTorch (<https://pytorch.org/>).

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Competing interests

Alibaba Group has filed for patent protection (CN116993663A; the other application number is not currently in the public domain) for work related to the methods of FLL detection and diagnosis. W.L., Yuan Gao, K.Y. and L.Z. are employees of Alibaba Group and own stock as part of their employment. S.O. is co-founder and non-executive director of Hypervision Surgical, co-founder of Hologen and advisor to Proximie, Elaitra and neuRealities. The other authors declare no competing interests.

Additional information

Extended data is available for this paper at <https://doi.org/10.1038/s41591-026-04589-y>.

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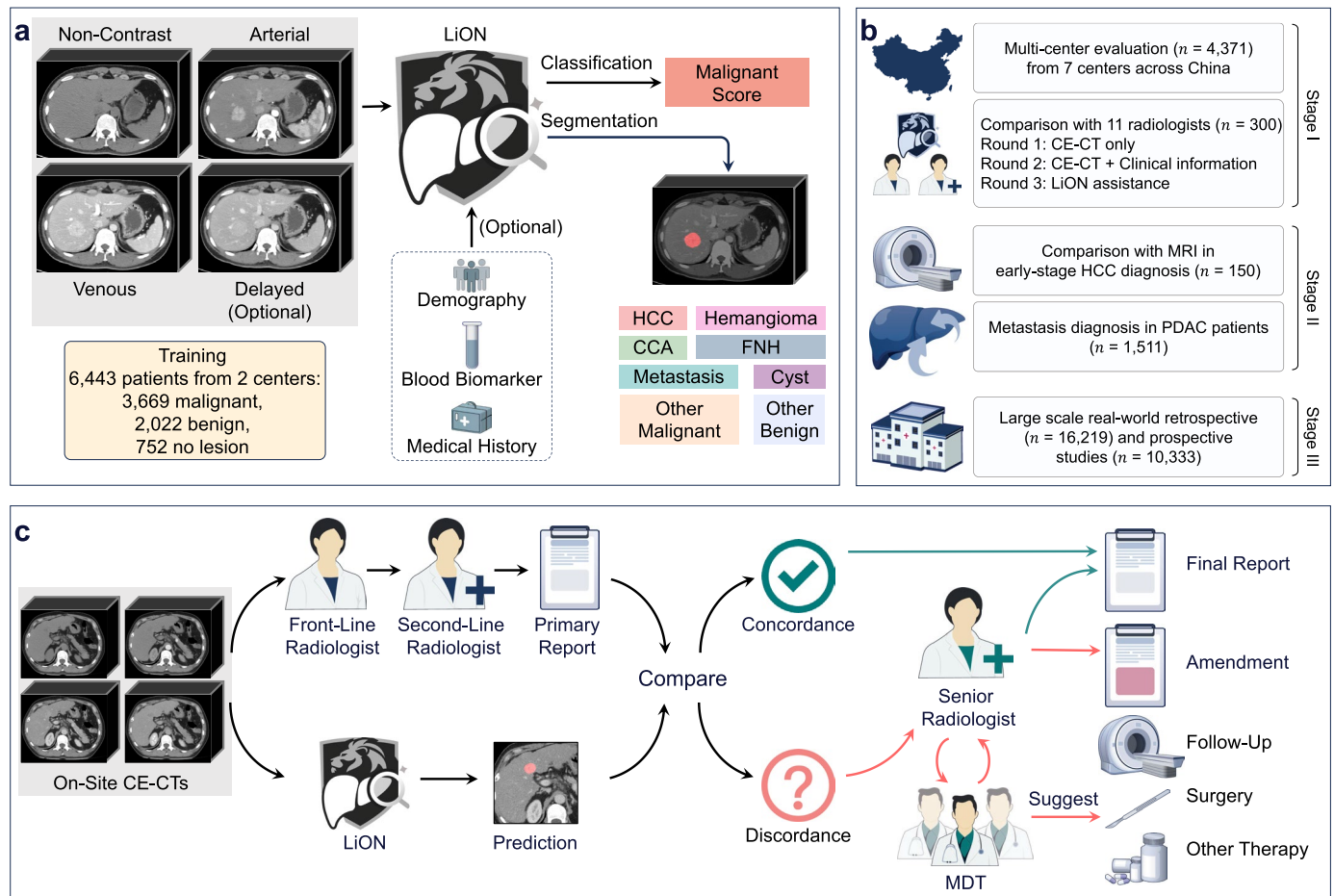
Author contributions

Conceptualization: S.O., M.A.Z., K.Y., Qi Zhang, Y. Hou, T.L., L.Z. and Y.S. Data curation: X.H., X.Y., C.Z., J.Z., K.C., Q.L., Y. Hong, Yingshan Gao, Qianqian Zhang, J.Q. and S.Z. Formal analysis: X.Z., C.L., X.H., M.S. and Y.S. Funding acquisition: C.L., K.C., Qianqian Zhang, T.J., O.H., M.A.Z., Qi Zhang, Y. Hou, T.L. and Y.S. Investigation: X.Z., C.L., X.H., X.Y., C.Z.,

Extended Data Table 1 | Additional cohort characteristics

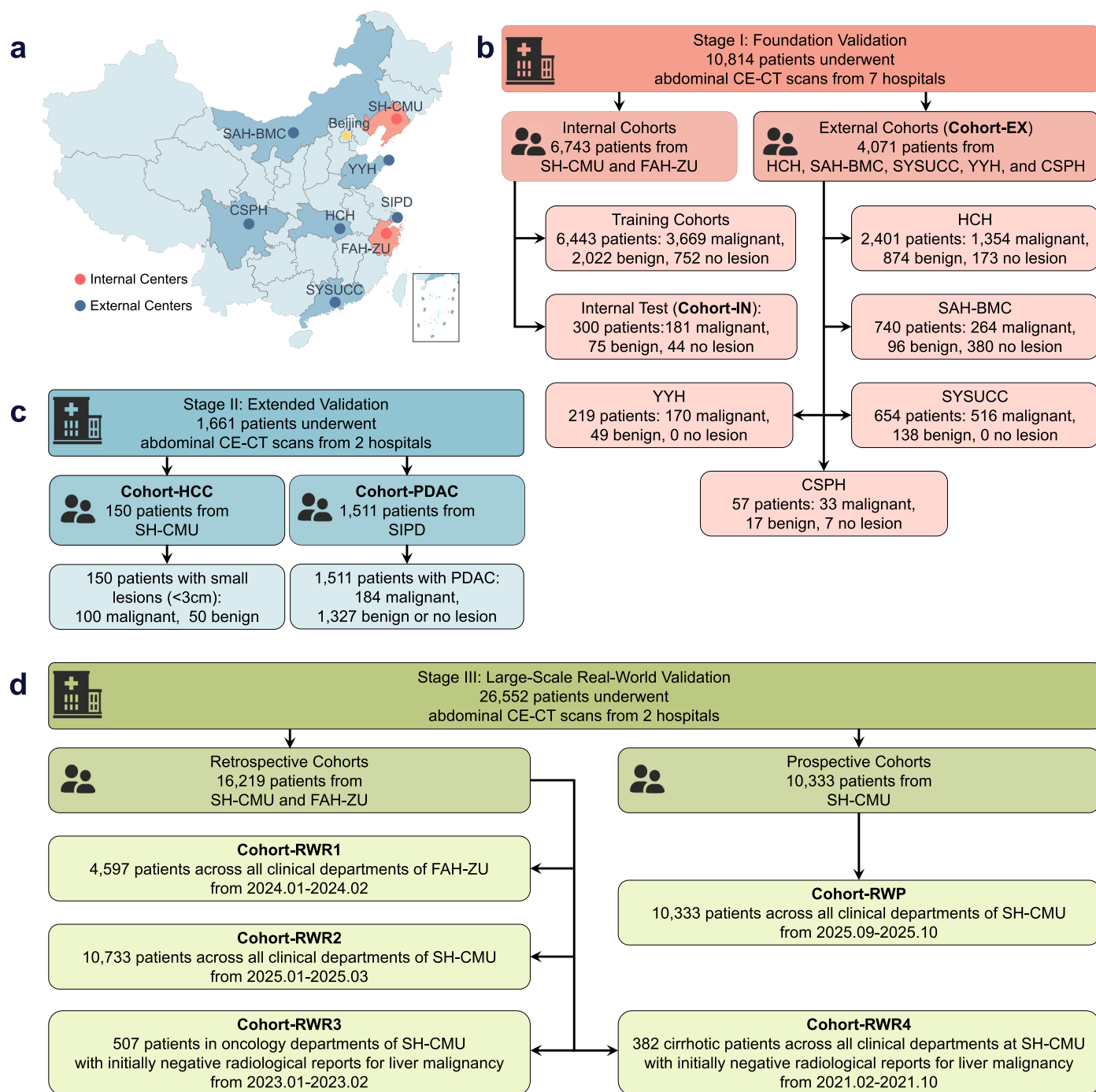
Stages		Stage I: foundation validation				Stage II: extended validation				Stage III: large-scale real-world validation			
Study phase		Training		Internal validation		External validation		Extended validation		Real-world retrospective		Real-world prospective	
Internal / External		Internal		External		Internal	External	Internal	External	Internal		Internal	
Cohort abbreviation			Cohort-IN	Cohort-EX				Cohort-HCC	Cohort-PDAC	Cohort-RWR1	Cohort-RWR2	Cohort-RWR3	Cohort-RWR4
Institute		SH-CMU	FAH-ZU	SH-CMU, FAH-ZU	HCH	SAH-BMC	SYSUCC	YTH	CSPH	SH-CMU	FAH-ZU	SH-CMU	SH-CMU
Total cohort size		2657	3786	300	2401	740	654	219	57	150	1511	10733	382
Extrahepatic malignancy		440 (16.6%)	832 (22.0%)	62 (20.7%)	972 (40.5%)	312 (42.2%)	47 (7.2%)	21 (9.6%)	35 (61.4%)	89 (59.3%)	1511 (100.0%)	—	—
Cirrhosis		354 (13.3%)	546 (14.4%)	100 (33.3%)	413 (17.2%)	115 (15.5%)	331 (50.6%)	110 (50.2%)	25 (43.9%)	14 (9.3%)	—	188 (4.1%)	382 (100.0%)
Thickness [1-3] (mm)		873 (32.9%)	1137 (30.0%)	65 (21.7%)	158 (6.6%)	23 (3.1%)	0 (0.0%)	0 (0.0%)	38 (66.7%)	29 (19.3%)	348 (23.0%)	199 (4.3%)	62 (12.2%)
Thickness (3-5] (mm)		1784 (671%)	2649 (70.0%)	235 (78.3%)	2243 (93.4%)	717 (96.9%)	654 (100.0%)	219 (100.0%)	19 (33.3%)	121 (80.7%)	1163 (77.0%)	4398 (95.7%)	9274 (86.4%)
Dates		2016-01 / 2021-12	2015-01 / 2023-09	2015-01 / 2023-09	2014-04 / 2025-06	2019-01 / 2019-10	2007-03 / 2021-03	2018-10 / 2019-04	2021-02 / 2025-02	2017-07 / 2024-10	2015-01 / 2021-12	2024-01 / 2024-02	2025-01 / 2025-03
Examination type		C+A+P: 123 (4.6%) A: 761 (28.6%) Liver: 22 (0.8%) C-A: 1663 (62.6%) Ves: 88 (3.3%)	C+A+P: 1458 (38.5%) A: 757 (20.0%) Pan: 655 (17.3%) Liver: 790 (20.9%) Sto: 14 (0.4%) C: 26 (1.7%) Ves: 86 (2.3%)	C+A+P: 16 (5.3%) A: 101 (33.7%) Liver: 6 (2.0%) C+A: 172 (57.3%) Ves: 5 (1.7%)	C+A+P: 1377 (57.4%) A: 819 (34.1%) Liver: 74 (3.1%) Kid+Adr: 5 (0.2%) Sto: 22 (0.9%) C: 42 (1.7%) Ves: 62 (2.6%)	C+A+P: 20 (2.7%) A: 720 (97.3%) Liver: 74 (3.1%) Kid+Adr: 5 (0.2%) Sto: 22 (0.9%) C: 42 (1.7%) Ves: 62 (2.6%)	C+A+P: 349 (53.4%) A: 158 (22.6%) Liver: 87 (13.3%) C: 49 (7.5%) Ves: 21 (3.2%)	C+A: 61 (27.9%) A: 158 (72.1%) Liver: 87 (13.3%) Head+C+A: 4 (7.0%)	C+A: 11 (19.3%) A: 42 (73.7%) Head+C+A: 4 (7.0%)	C+A: 14 (9.3%) A: 13 (8.7%) Liver: 123 (82.0%)	A+P: 119 (7.9%) A: 752 (49.8%) Pan: 640 (42.3%)	A: 3859 (83.9%) C-A: 223 (4.9%) Liver: 285 (6.2%) Ves: 230 (5.0%)	A+P: 6854 (63.9%) A: 750 (7.0%) Liver: 1019 (9.5%) Uro: 1082 (10.1%) Kid: 299 (2.8%) Adr: 418 (3.9%) Pan: 243 (2.3%) Ves: 68 (0.6%)
KVP (kV)		120: 1812 (68.2%) 140: 638 (24.0%) 100: 207 (7.8%)	140: 35 (0.9%) 120: 3691 (97.5%) 100: 55 (1.5%) 90: 5 (0.1%)	120: 142 (47.3%) 140: 131 (43.7%) 100: 27 (9.0%)	120: 2329 (97.0%) 100: 38 (1.6%) 80: 34 (1.4%)	120: 604 (81.6%) 100: 133 (18.0%) 140: 3 (0.4%)	150: 14 (2.1%) 140: 20 (3.1%) 120: 433 (66.2%) 100: 151 (23.1%) 90: 36 (5.5%)	120: 201 (91.8%) 100: 18 (8.2%)	120: 54 (94.7%) 100: 3 (5.3%)	140: 18 (12.0%) 120: 125 (83.3%) 100: 7 (4.7%)	120: 1511 (100.0%)	140: 117 (23.1%) 120: 344 (67.8%) 100: 46 (9.1%)	140: 608 (5.7%) 120: 8385 (78.1%) 100: 1740 (16.2%)

A hyphen (—) indicates not applicable. Institutes: SH-CMU Shengjing Hospital of China Medical University; FAH-ZU The First Affiliated Hospital, Zhejiang University School of Medicine; HCH Hubei Cancer Hospital; SAH-BMC The Second Affiliated Hospital of Baotou Medical College, Inner Mongolia University of Science and Technology; SYSUCC Sun Yat-sen University Cancer Center; YYH Yantai Yuhuangding Hospital; CSPH Chengdu Sixth People's Hospital; SIFD Shanghai Institution of Pancreatic Diseases. Examination Types: C+A+P Chest+abdomen+pelvis; C+A Chest+abdomen; A+P Abdomen+Pelvis; A Abdomen; C Chest; Kid Kidney; Adr Adrenal; Pan Pancreas; Sto Stomach; Uro Urography; Ves Vessel & other specialized scans. Scanner Manufacturers: Phi Philips; Sie Siemens; Tos Toshiba; NMS Neusoft Medical Systems; UJH United Imaging Healthcare; Min Minfound.



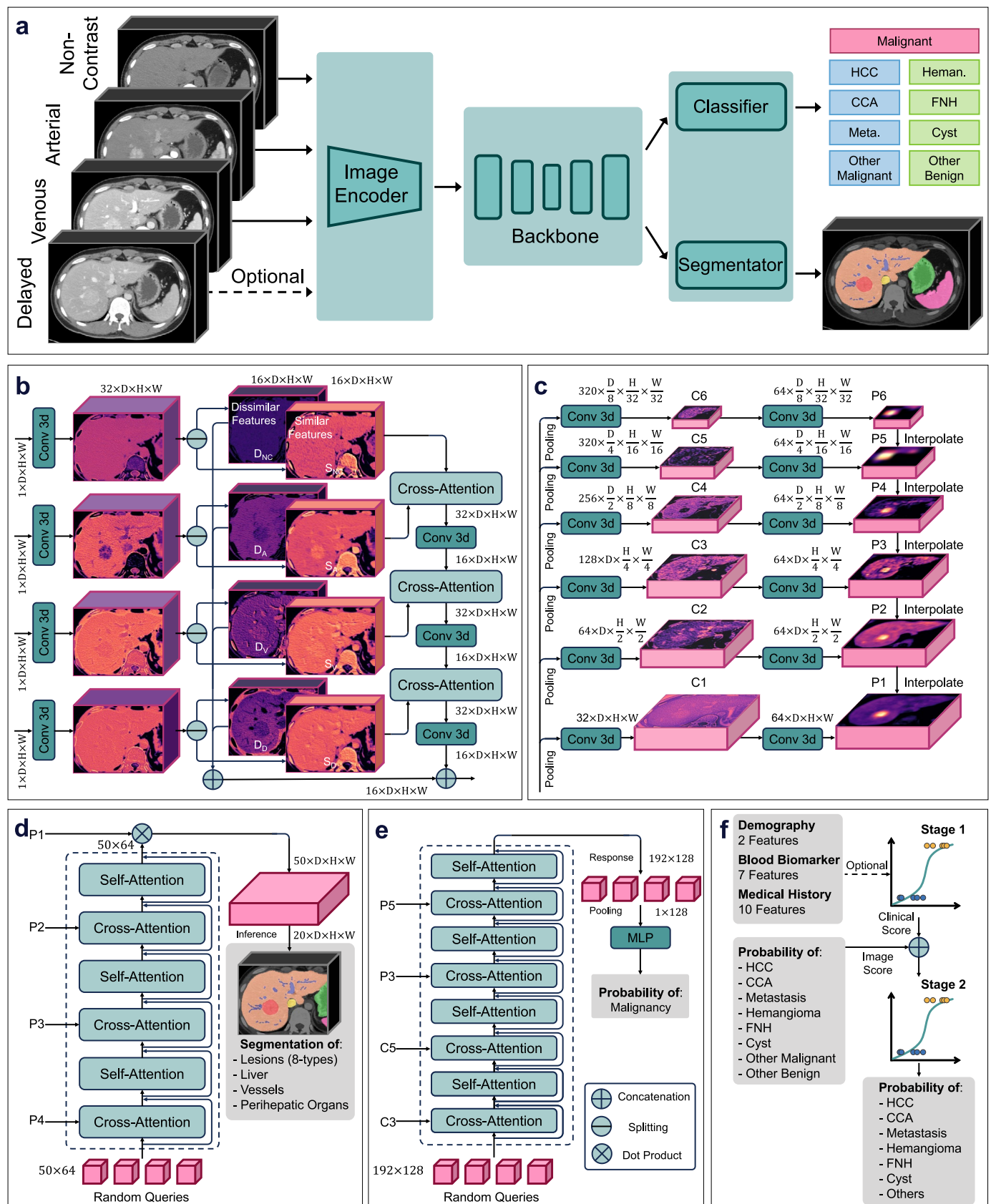
Extended Data Fig. 1 | The LiON AI system: development, validation, and transformation of the clinical pathway. a, Model development. LiON analyzes CE-CTs with adaptable input for delayed-phase images, predicting patient-level malignancy scores and segmenting eight types of liver lesions. When available, clinical data integration further enhances patient-level diagnostic prediction. **b**, Model evaluation. LiON underwent comprehensive validation through three

stages including foundational validation, extended validation, and large-scale real-world validation. **c**, AI-human collaboration workflow in prospective study. LiON integrated into clinical decision-making processes. *CE-CT* Contrast-Enhanced Computed Tomography; *PDAC* Pancreatic Ductal Adenocarcinoma; *MDT* Multidisciplinary Team.



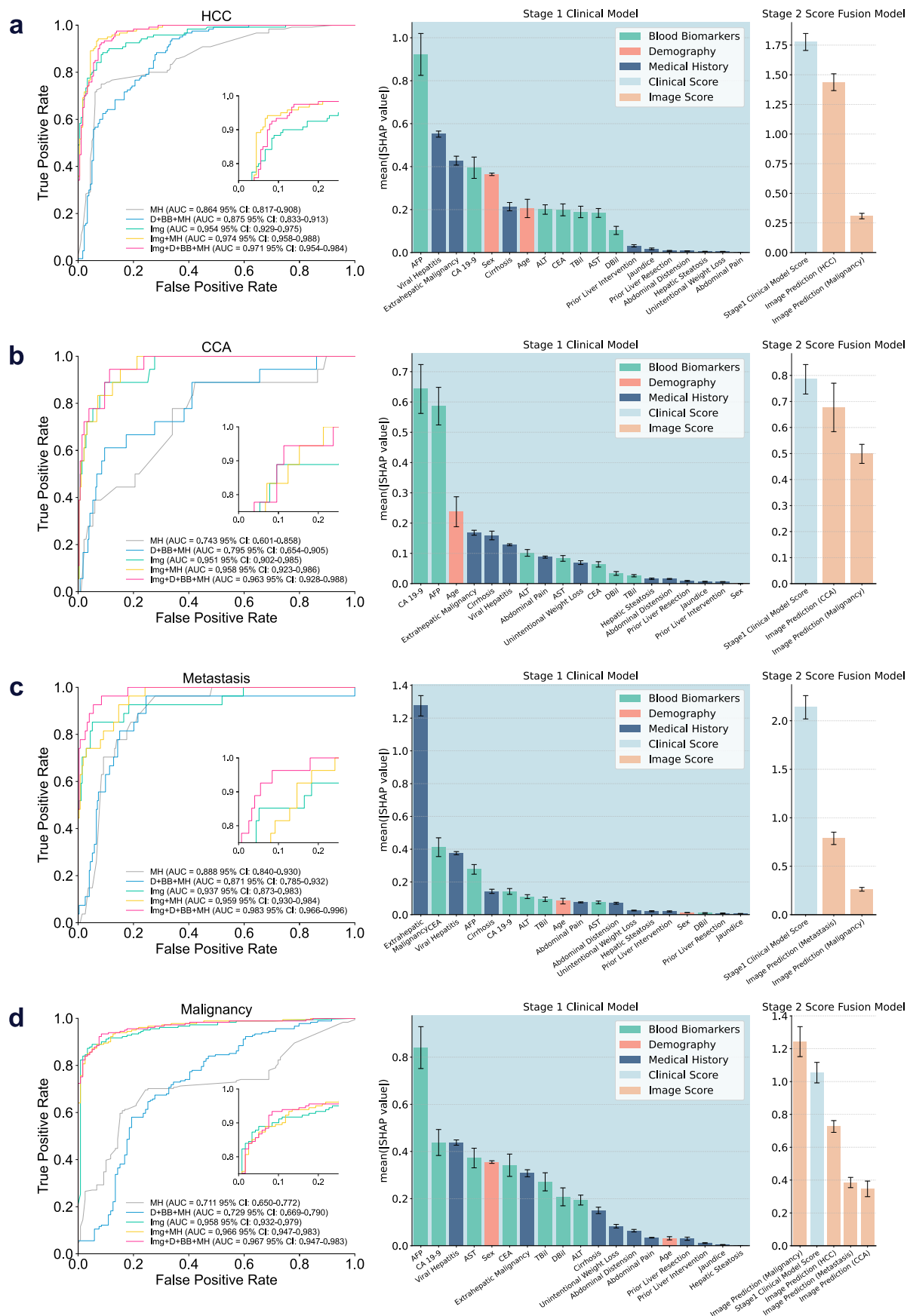
Extended Data Fig. 2 | Cohort distribution. **a**, Geographic distribution of participating centers in China. **b**, Stage I: foundation validation. **c**, Stage II: extended validation. **d**, Stage III: large-scale real-world validation. *CE-CT* Contrast-Enhanced Computed Tomography; *SH-CMU* Shengjing Hospital of China Medical University; *FAH-ZU* The First Affiliated Hospital, Zhejiang

University School of Medicine; *HCH* Hubei Cancer Hospital; *SAH-BMC* The Second Affiliated Hospital of Baotou Medical College, Inner Mongolia University of Science and Technology; *SYSUCC* Sun Yat-sen University Cancer Center; *YYH* Yantai Yuhuangding Hospital; *CSPH* Chengdu Sixth People's Hospital; *SIPD* Shanghai Institution of Pancreatic Diseases.



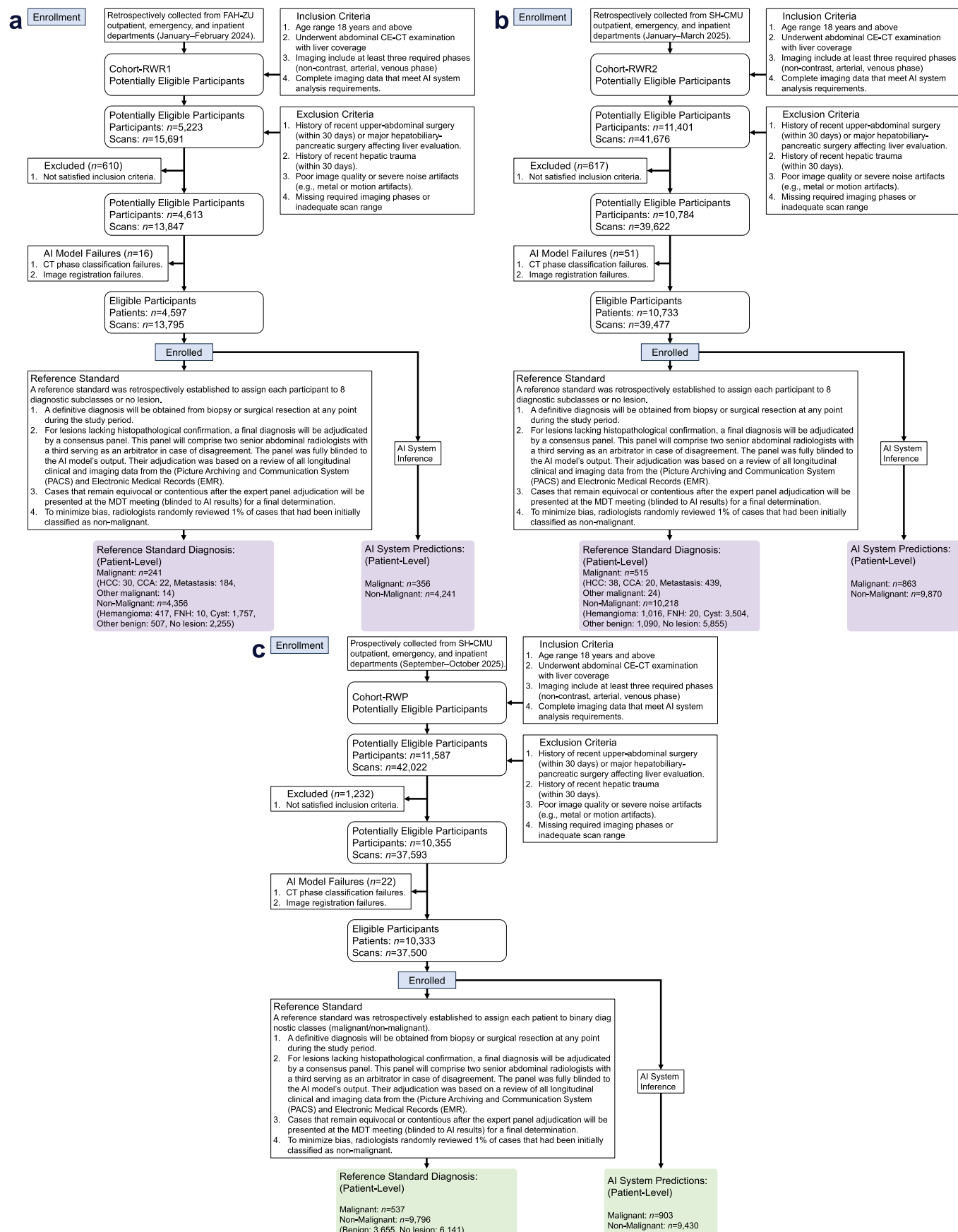
Extended Data Fig. 3 | Architecture of LiON. a, Pipeline overview: The deep learning framework consists of four components (image encoder, backbone, classifier, and segmentator) that achieve end-to-end liver diagnosis. **b**, Image encoder architecture accepting three required phases (non-contrast, arterial, and venous) with optional delayed phase input. **c**, Backbone architecture

utilizing feature pyramid for multi-scale feature construction. **d**, Segmentator design employing learnable queries and dense cross-attention for instance segmentation. **e**, Classifier structure using learnable queries and dense cross-attention for patient-level prediction. **f**, Optional clinical features enhance prediction through two-stage score-based logistic regression.



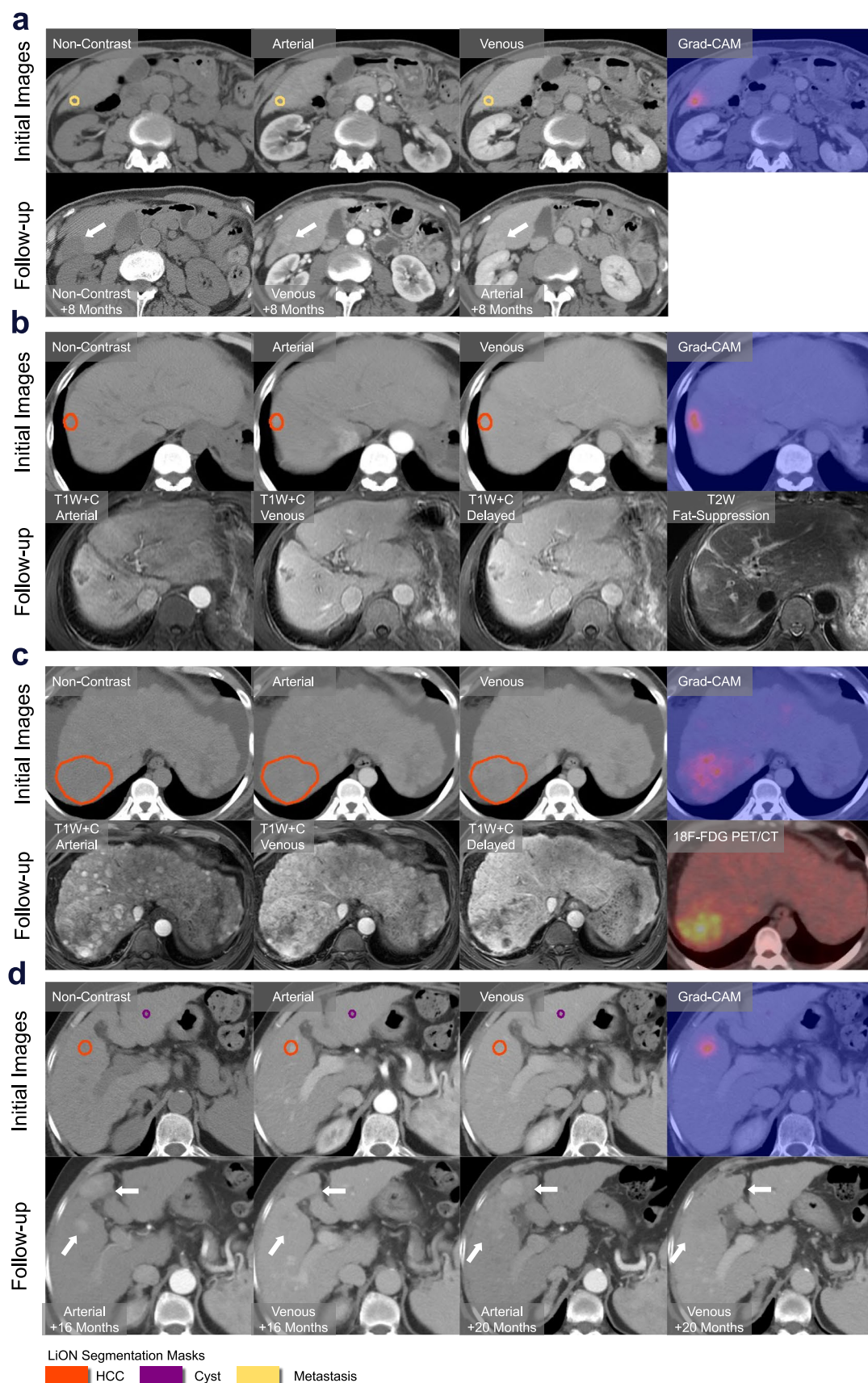
Extended Data Fig. 4 | Multi-modal integration of clinical information. a-d. Receiver operating characteristic curves showing diagnostic performance with integrated clinical features and SHAP-based feature importance analysis on Cohort-IN ($n = 300$ patients). Bars represent mean absolute SHAP values.

Error bars represent 95% confidence intervals: **a.** HCC, **b.** CCA, **c.** metastasis, and **d.** overall malignancy. *LiON* Liver Diagnosis Network; *HCC* Hepatocellular Carcinoma; *CCA* Intrahepatic Cholangiocarcinoma; *D* Demography; *BB* Blood Biomarkers; *MH* Medical History; *SHAP* SHapley Additive exPlanations.



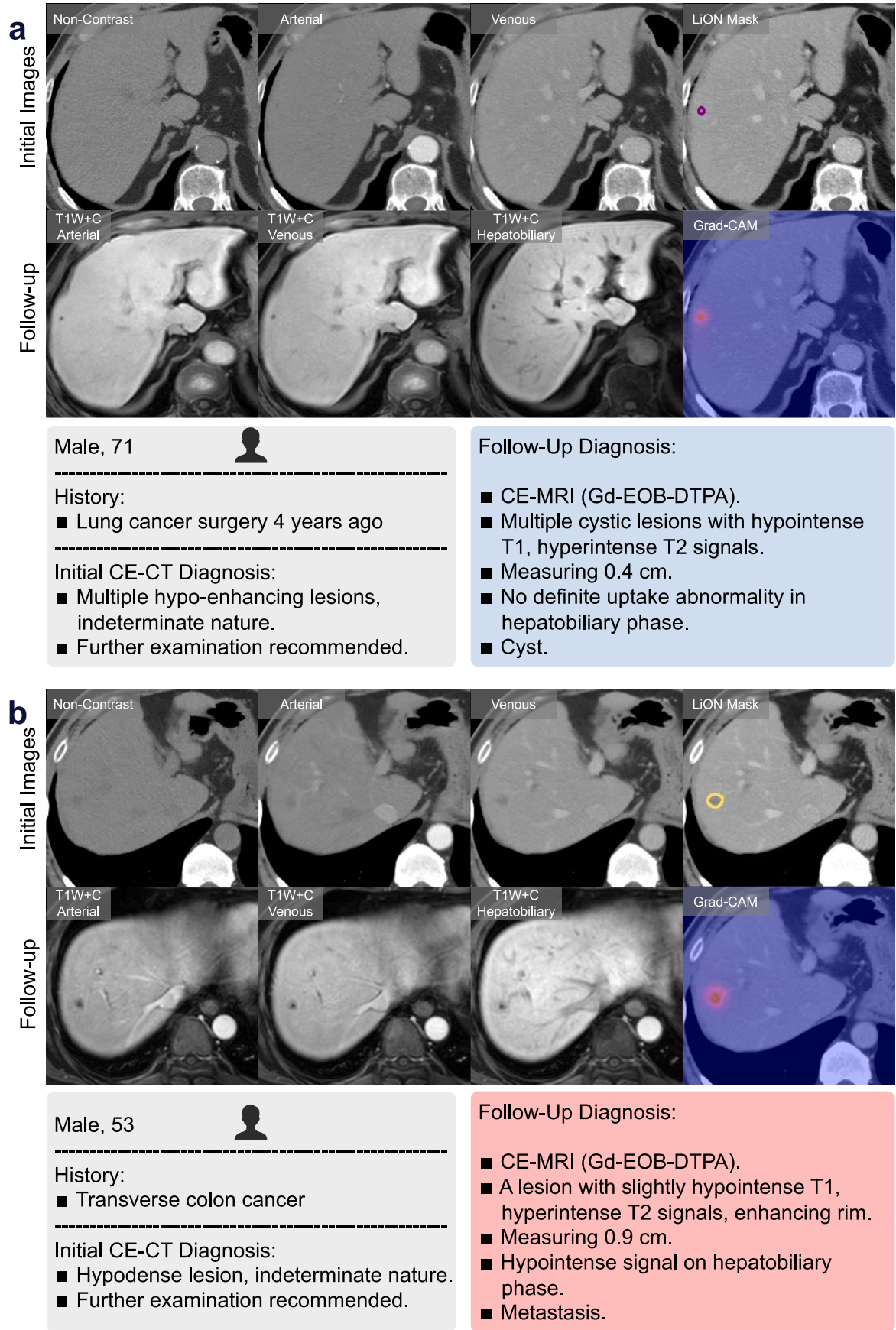
Extended Data Fig. 5 | Real-world validation workflow and reference standard establishment. **a**, Consecutive patients from retrospective cohort (Cohort-RWR1, $n=4,597$) from FAH-ZU. **b**, Consecutive patients from retrospective cohort (Cohort-RWR2, $n=10,733$) from SH-CMU. **c**, Consecutive patients from prospective

cohort (Cohort-RWP, $n=10,333$) from SH-CMU. FAH-ZU The First Affiliated Hospital, Zhejiang University School of Medicine; SH-CMU Shengjing Hospital of China Medical University.



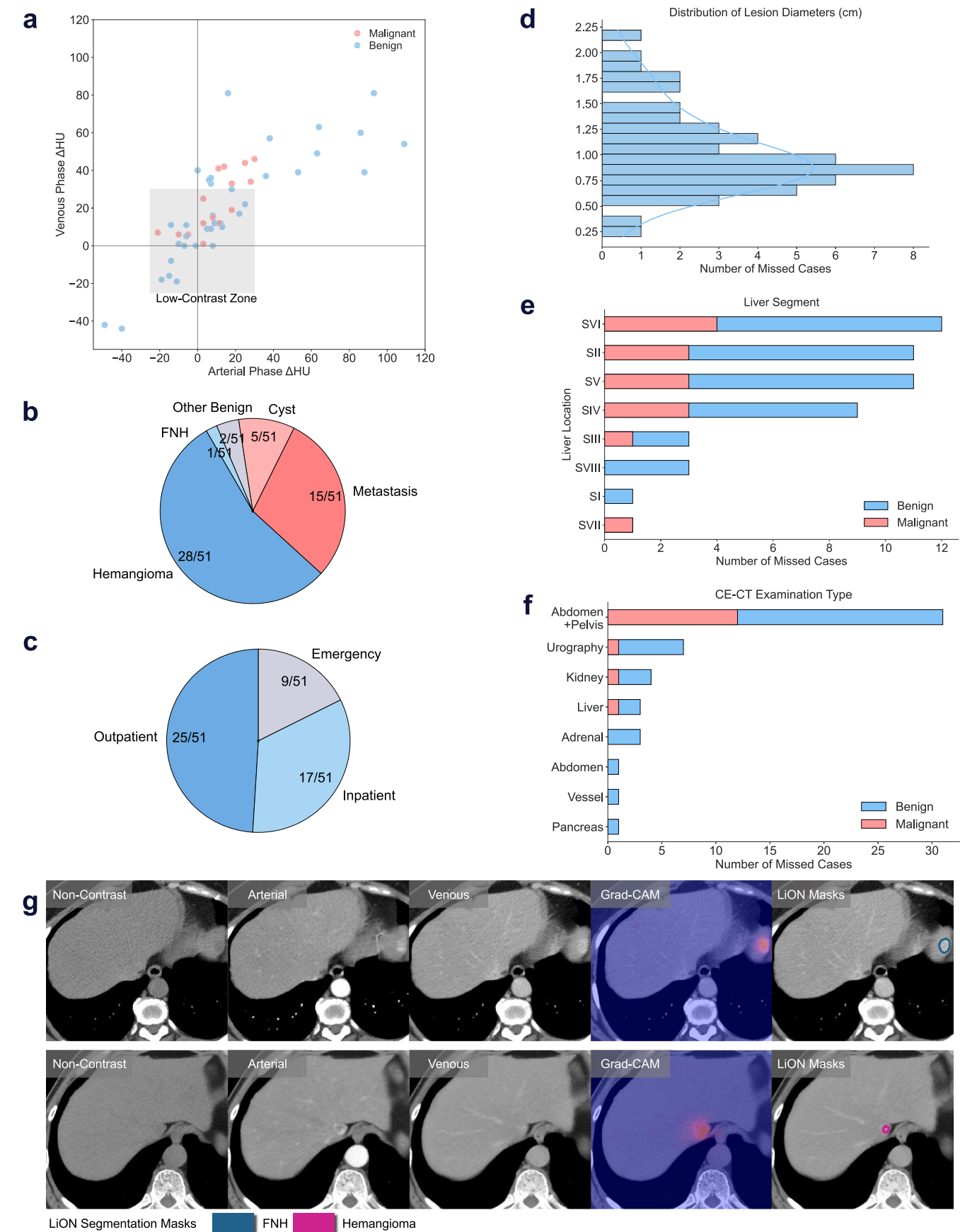
Extended Data Fig. 6 | Malignancies recalled in Cohort-RWR3 (oncology department) and Cohort-RWR4 (cirrhosis). **a**, Cohort-RWR3 case showing metastasis progression in a 70-year-old male during 8-month CE-CT follow-up after an initially non-malignant report. **b-d**, Cohort-RWR4 cases with initially non-malignant reports: **(b)**, 57-year-old male, confirmed by CE-MRI; **(c)**, 39-year-old

male with severe cirrhosis, confirmed by CE-MRI and ^{18}F -FDG PET/CT; **(d)**, 65-year-old female with severe cirrhosis, confirmed by follow-up imaging. *CE-CT* Contrast-Enhanced Computed tomography; *CE-MRI* contrast-enhanced magnetic resonance imaging; ^{18}F -FDG PET/CT ^{18}F -Fluorodeoxyglucose Positron Emission Tomography/Computed Tomography.



Extended Data Fig. 7 | Examples of initially indeterminate diagnosis (•) on CE-CTs in Cohort-RWP. a-b, Two abdominal CE-CT examinations with initially indeterminate radiology reports. Follow-up CE-MRI confirmed LiON's initial

diagnosis of (a). non-malignant cyst and (b). malignant metastasis, respectively. CE-MRI Contrast-Enhanced Magnetic Resonance Imaging; Gd-EOB-DTPA Gadolinium Ethoxybenzyl Diethylenetriamine Pentaacetic Acid.



Extended Data Fig. 8 | See next page for caption.

Extended Data Fig. 8 | Characteristics of initially overlooked lesions in Cohort-RWP. **a**, Distribution of enhancement patterns among overlooked lesions, predominantly characterized by low enhancement. **b**, Distribution of overlooked lesions by pathological type. **c**, Distribution of overlooked lesions across different clinical settings. **d**, Size distribution of overlooked lesions, measured

according to RECIST 1.1 criteria. **e**, Anatomical distribution of overlooked lesions by liver segments. **f**, Distribution of overlooked lesions by examination types. **g**, Demonstration of lesions overlooked in primary radiological reports and subsequently detected by LiON. *HU*/Hounsfield Units; *RECIST* Response Evaluation Criteria in Solid Tumors.

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☐	☒ A statement on whether measurements were taken from distinct samples or whether the same sample was measured repeatedly
☐	☒ The statistical test(s) used AND whether they are one- or two-sided <i>Only common tests should be described solely by name; describe more complex techniques in the Methods section.</i>
☒	☐ A description of all covariates tested
☒	☐ A description of any assumptions or corrections, such as tests of normality and adjustment for multiple comparisons
☐	☒ A full description of the statistical parameters including central tendency (e.g. means) or other basic estimates (e.g. regression coefficient) AND variation (e.g. standard deviation) or associated estimates of uncertainty (e.g. confidence intervals)
☐	☒ For null hypothesis testing, the test statistic (e.g. <i>F</i> , <i>t</i> , <i>r</i>) with confidence intervals, effect sizes, degrees of freedom and <i>P</i> value noted <i>Give P values as exact values whenever suitable.</i>
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☒	☐ Estimates of effect sizes (e.g. Cohen's <i>d</i> , Pearson's <i>r</i>), indicating how they were calculated

Our web collection on [statistics for biologists](#) contains articles on many of the points above.

Software and code

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Data collection	Dicom files were handled with the open source libraries Pydicom(https://pydicom.github.io/ , version 2.2.2), SimpleITK(https://simpleitk.org/ , version 2.0.2), and NiBabel(https://nipy.org/nibabel/ , version 3.2.1).
Data analysis	Experiments were conducted using Python 3.11.5 with PyTorch 2.1.0+cu118, CUDA 12.2, and other libraries including nnU-Net v1, SciPy 1.11.4, Torchvision 0.16.0+cu118, Scikit-learn 1.3.2, Pandas 2.1.4, NumPy 1.26.2, Scikit-image 0.25.2, SimpleITK 2.4.1, MedPy 0.5.2, Nibabel 5.3.2, and Matplotlib 3.8.2. The full source code for LiON cannot be publicly released due to dependencies on proprietary internal infrastructure (e.g., phase classification, image registration, and rendering) and pending patent protection (CN116993663A). All experiments and implementation details are described in the Methods and Supplementary Materials to support replication using non-proprietary libraries. The key algorithmic components and inference logic for FFL detection and diagnosis are available in the open-source repository (https://github.com/alibaba-damo-academy/pixel-lesion-patient-network/). The framework was implemented in PyTorch (https://pytorch.org/).

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Due to ethical restrictions protecting patient confidentiality, the datasets analyzed in this study are not publicly available. Qualified researchers may request access to anonymized individual-level data and related clinical documentation by contacting corresponding authors (Q.Z. qi.zhang@zju.edu.cn, Y.Hou houy2@sj-hospital.org, T.L. liangtingbo@zju.edu.cn, and Y.S 18940259980@163.com). Requests will be evaluated by the relevant institutional data-governance committees based on compliance with institutional and regulatory requirements. Access will require approval from the relevant institutional review board where applicable and completion of a data-use agreement. Requests will be processed within a maximum of 6 weeks.

Research involving human participants, their data, or biological material

Policy information about studies with [human participants or human data](#). See also policy information about [sex, gender \(identity/presentation\), and sexual orientation](#) and [race, ethnicity and racism](#).

Reporting on sex and gender

Sex was recorded based on government-issued IDs across all datasets, including the internal training and validation cohorts, external multi-center validation cohorts, extended validation cohorts, retrospective large-scale validation cohorts, and the real-world prospective large-scale validation cohort. Gender information was not collected from the patients.

Reporting on race, ethnicity, or other socially relevant groupings

Race, ethnicity, and other socially relevant groupings were not collected from the patients and were not factors in model implementation or deployment.

Population characteristics

LiON was trained on 6,443 patients from two major general medical centers Shengjing Hospital of China Medical University (SH-CMU) and the First Affiliated Hospital, Zhejiang University School of Medicine (FAH-ZU). The institutional CE-CT protocols included standard three phases (non-contrast, arterial, and venous), while the delayed phase was performed according to specific examination requirements. The training cohorts included patients over 18 years of age who had undergone complete multiphasic CE-CT examinations, with malignancy confirmed by pathological analysis and benign lesions confirmed either by pathology or by a minimum one-year imaging follow-up.

Real-world retrospective study: For the real-world retrospective study, we deployed LiON across four cohorts. The primary validation was conducted on 15,330 consecutive patients from two centers (Cohort RWR1, n = 4,597; Cohort-RWR2, n = 10,733). Additionally, we established two dedicated screening cohorts from SH-CMU: a general outpatient cohort (Cohort-RWR3, n = 507) and a high-risk cirrhotic cohort (Cohort-RWR4, n = 382).

Real-world prospective study: To prospectively evaluate LiON's real-world clinical impact, we deployed it in an interventional study at SH-CMU (Cohort-RWP), analyzing 10,333 consecutive patients between September and October 2025. The system ran in parallel with the standard two-tier clinical workflow (first-line radiologist reporting with subsequent senior review). A dedicated senior radiologist then performed adjudication of any discordances between AI findings and primary radiological reports. Discrepancies were escalated to a consensus panel or MDT for final review and potential clinical intervention.

Recruitment

The inclusion criteria encompassed all CE-CT scans involving the liver, including abdominal scans with different anatomical coverage (e.g., pelvis, upper/lower abdomen) and organ-specific protocols (e.g., pancreatic, urographic scans). Patients with major liver surgery and acute injury were excluded.

This study utilized a multicenter dataset to develop and validate LiON (Extended Data Fig. 2) for liver diagnosis using CE-CT. In total, 39,027 patients with 139,801 CT scans were included. Detailed patient characteristics and the nomenclature of each cohort are provided in Table 1, Extended Data Table 1 and Supplementary Table S1, respectively. To ensure broad geographical representation and minimize regional bias, we collected data from 6 regions across China (Northeast, East, South, North, Central, and Southwest) through a consortium of seven major medical centers (Extended Data Fig. 3). The external centers included two major oncology specialist hospitals, Hubei Cancer Hospital (HCH) and Sun Yat-sen University Cancer Center (SYSUCC), along with three general hospitals: The Second Affiliated Hospital of Baotou Medical College (SAH-BMC), Yantai Yuhuangding Hospital (YYH), and Chengdu Sixth People's Hospital (CSPH). The PDAC data for hepatic metastasis diagnosis were obtained from Shanghai Institution of Pancreatic Diseases (SIPD).

For comprehensive multi-center validation, two major cohorts were included: a held-out internal validation set (Cohort-IN, n=300) of randomly selected cases from the internal centers: SH-CMU (n=124) and FAH-ZU (n=176), and an external test set (Cohort-EX, n=4,071) collected from five external centers: HCH (n=2,401), SAH-BMC (n=740), SYSUCC (n=654), YYH (n=219), and CSPH (n=57). To assess performance in clinically challenging scenarios, we included Cohort-HCC (n=150), a small-lesion cohort comprising patients from SH-CMU with lesions ≤ 3 cm in diameter, and Cohort-PDAC (n=1,511) from SIPD. Cohort-PDAC comprised 546 patients with resectable pancreatic cancer, 123 with borderline resectable pancreatic cancer, 573 with locally advanced pancreatic cancer, and 269 with MPC, including 184 with hepatic metastases. Hepatic metastasis was confirmed through histopathological biopsy or, when biopsy was contraindicated or unfeasible in advanced PDAC, through comprehensive MDT evaluation of clinical evidence.

For real-world performance assessment, we established testing cohorts from internal centers (FAH-ZU and SH-CMU),

encompassing 26,552 patients over 18 years who underwent CE-CT with liver coverage for various clinical indications, representing diverse clinical scenarios (outpatient, emergency, and inpatient). The inclusion criteria encompassed all CE-CT scans involving the liver, including abdominal scans with different anatomical coverage (e.g., pelvis, upper/lower abdomen); and organ-specific protocols (e.g., pancreatic, urographic scans). Patients with major liver surgery and acute injury were excluded. The detailed process for establishing reference standards is described in Extended Data Fig. 5.

Ethics oversight

All research activities adhered to the principles of the Declaration of Helsinki and institutional guidelines. This study was conducted under institutional review board approval from all participating centers (SH-CMU (2024PS932K), FAH-ZU ([2022B] IIT Ethics Approval No.1007, [2025B] IIT Ethics Approval No.0423), HCH (LLHBCH2025YN-040), SAH-BMC (2024-ZX-051), SYSUCC (B2025-290-01), YYH (2024-624), CSPH (2025-L-007), and SIPD (CHEC2022-069)). The prospective component was approved by the institutional review board of SH-CMU and registered at ClinicalTrials.gov (NCT07153783), and adhered to the CONSORT-AI guidelines. The ethics committees waived informed consent requirements for retrospectively collected CT images and clinical data. Written informed consent was obtained from patients whose clinical information, CE-CT examinations, and follow-up records were prospectively collected.

Note that full information on the approval of the study protocol must also be provided in the manuscript.

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Sample size

This study utilized a multi-center dataset to develop and validate LiON (Extended Data Fig. 1) for liver diagnosis using CE-CT, in total, 39,027 patients with 139,801 CT scans were collected. Detailed patient characteristics and the nomenclature of each cohort are provided in Table 1, Extended Data Table 1 and Supplementary Table S1, respectively. The training cohorts, comprising 6,443 patients from two general medical centers (SH-CMU and FAH-ZU), were used for model development. The study included patients over 18 years of age who had confirmed pathological diagnosis for malignancy and complete multiphase CE-CT examinations. Of these, the primary training dataset included 3,669 malignant cases, 2,022 benign cases, and 752 no-lesion cases. The dataset encompassed diverse lesion pathologies. Common malignancy diagnoses included hepatocellular carcinoma (HCC), intrahepatic cholangiocarcinoma (CCA), and metastasis, while benign lesions primarily consisted of hemangioma, focal nodular hyperplasia (FNH), and cysts. Lesion volumes differed between centers, with SH-CMU showing larger mean volumes (48.87±192.61 cm³ versus 25.68±102.11 cm³). To address the challenge of varying scanning protocols across different centers, we collected both 3-phase (non-contrast, arterial, venous, without delayed phase) and 4-phase (with delayed phase) CE-CT scans. The training set included 2,567 3-phase and 3,876 4-phase scans (average 3.60 scans per patient), with HCH predominantly contributing 3-phase scans because of variations in institutional examination protocols. To assess performance in clinically challenging scenarios, we included Cohort-HCC (n=150), a small-lesion cohort comprising patients from SH-CMU with lesions ≤ 3 cm in diameter from SH-CMU, and Cohort-PDAC (n=1,511) from Shanghai Institution of Pancreatic Diseases (SIPD). Cohort-PDAC (n=1,511), comprised 547 resectable pancreatic cancer, 123 borderline resectable pancreatic cancer, 573 locally advanced pancreatic cancer, and 269 metastatic pancreatic cancer, including 194 with hepatic metastases. Hepatic metastasis was confirmed through histopathological biopsy or, when biopsy was contraindicated or unfeasible in advanced pancreatic ductal adenocarcinoma (PDAC), through comprehensive MDT evaluation of clinical evidence. For real-world performance assessment, we established testing cohorts from internal centers (FAH-ZU and SH-CMU), encompassing 26,552 patients over 18 years who underwent CE-CT with liver coverage for various clinical indications, representing diverse clinical scenarios (outpatient, emergency, and inpatient).

For the retrospective development and validation cohorts, no formal sample size calculation was performed. Sample sizes were determined by the availability of eligible cases for model development and independent validation. Their adequacy was supported by training-size sensitivity analyses in the supplementary, which showed that model performance approached a plateau as increasing proportions of the training data were used. Independent validation cohorts were then used to assess generalizability across different clinical settings and lesion distributions.

For the single-arm trial, the sample size was calculated to provide 90% statistical power (1 - β) to test the one-sided superiority hypothesis (H1: AUC > 0.900) against a benchmark of 0.900, at a significance level of α = 0.025. Assuming a malignancy prevalence of ≈5.0%, an expected AUC of 0.950 based on real-world retrospective study, and a 20% rate of data loss or attrition, a target enrollment of approximately 3,951 participants was required (using PASS 2022, NCSS, LLC, Kaysville, UT, USA). We consecutively enrolled all eligible patients during the 2-month study period, anticipating approximately 10,000 participants based on institutional patient volume. This larger enrollment was chosen to improve precision in estimating the primary AUC and to support exploratory subgroup analyses. Finally, 10,333 participants were enrolled in the full analysis set.

Data exclusions

History of recent upper-abdominal surgery (within 30 days) or major hepatobiliary-pancreatic surgery affecting liver evaluation (e.g., liver transplantation or Whipple procedure);
History of recent hepatic trauma (within 30 days);
Poor image quality or severe noise artifacts (e.g., metal or motion artifacts);
Missing required imaging phases (required at least non-contrast, arterial, and venous phases) or inadequate scan range (e.g., lower-abdomen CT such as pelvic or rectal scans not covering the liver).

Replication

The performance of the classification was evaluated using the AUC, sensitivity, specificity, PPV, NPV and balanced accuracy. Confidence intervals were calculated based on 2,000 bootstrap replications of the corresponding data.

All attempts at replication were successful. The AI model demonstrated consistent performance across internal and external centers, regardless of the equipment manufacturer (e.g., GE, Philips, Toshiba, Siemens, UIH) or examination type (chest-abdomen, abdomen, liver, and other specialized scans with liver coverage). Furthermore, the model's performance was consistent between the large-scale retrospective and prospective studies.

Randomization	The internal dataset was randomly split into training, validation, and testing sets. Within the training cohort, patients were randomly assigned to training and validation folds during cross-validation.
Blinding	In the reader study, readers were blinded to the AI results and clinical information in round 1, and remained blinded to the AI results in round 2. Readers were blinded to the data source, the disease prevalence, and other readers' assessments. In the prospective study, the multidisciplinary team was blinded to the AI diagnosis.

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Clinical data

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Clinical trial registration	ClinicalTrials.gov registration: NCT07153783
Study protocol	https://clinicaltrials.gov/study/NCT07153783
Data collection	For the real-world retrospective study, 10,733 consecutive patients were included from SH-CMU (January 1 to March 19, 2025), and 4,597 consecutive patients were included from the outpatient, emergency, and inpatient departments of The First Affiliated Hospital, Zhejiang University School of Medicine (FAH-ZU) (January 1 to February 29, 2024). For the single-arm trial, 10,333 consecutive patients were enrolled from the outpatient, emergency, and inpatient departments of Shengjing Hospital of China Medical University (SH-CMU) between September 1 and October 29, 2025. Participants were recruited consecutively from patients undergoing clinically indicated contrast-enhanced CT as part of routine care at SH-CMU. Self-selection including CT-specific referral patterns, differences in lesion spectrum and pretest malignancy risk, and exclusion of patients unsuitable for contrast-enhanced CT. Participants were recruited consecutively from patients undergoing clinically indicated contrast-enhanced CT as part of routine care at SH-CMU. Potential sources of bias include CT-specific referral patterns, differences in lesion spectrum and pretest malignancy risk, and exclusion of patients unsuitable for contrast-enhanced CT.
Outcomes	For the single-arm trial, the primary endpoint was the AUC of the LiON system for malignancy diagnosis. The 95% CI for the AUC was calculated using 2,000 bootstrap replications of the corresponding data. The success criterion was met if the lower bound of the 95% CI was greater than 0.900. Secondary performance metrics included sensitivity, specificity, PPV, and NPV. CIs for these metrics were estimated using 2,000 bootstrap replications. Inter-observer agreement was evaluated using Cohen's κ . All secondary hypothesis tests were two-sided with a significance threshold of $p < 0.05$. Pre-specified subgroup analyses were performed on lesions with indeterminate or borderline diagnoses to evaluate performance and agreement in difficult cases. Missing data were handled according to the complete-case principle for the primary analysis.

Plants

Seed stocks	n/a
Novel plant genotypes	n/a
Authentication	n/a